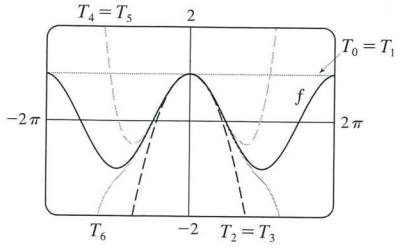
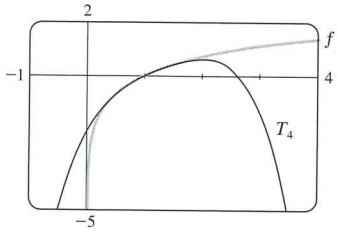
(c) As n increases, $T_n(x)$ is a good approximation to $f(x)$ on a larger and larger interval.



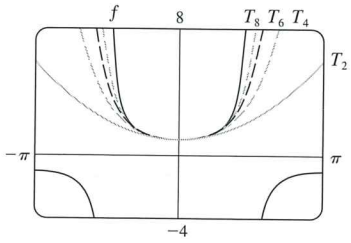
3. $(x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4$



5. $\frac{1}{2} + \frac{\sqrt{3}}{2}\left(x - \frac{\pi}{6}\right) - \frac{1}{4}\left(x - \frac{\pi}{6}\right)^2 - \frac{\sqrt{3}}{12}\left(x - \frac{\pi}{6}\right)^3$

7. $x + \frac{1}{3}x^3$ **9.** $x + x^2 + \frac{1}{3}x^3$

11. $T_8(x) = 1 + \frac{1}{2}x^2 + \frac{5}{24}x^4 + \frac{61}{720}x^6 + \frac{277}{8064}x^8$



13. (a) $2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$ (b) 1.5625×10^{-5}

15. (a) $\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\left(x - \frac{\pi}{4}\right) - \frac{1}{2\sqrt{2}}\left(x - \frac{\pi}{4}\right)^2 - \frac{1}{6\sqrt{2}}\left(x - \frac{\pi}{4}\right)^3 + \frac{1}{24\sqrt{2}}\left(x - \frac{\pi}{4}\right)^4 + \frac{1}{120\sqrt{2}}\left(x - \frac{\pi}{4}\right)^5$

(b) 0.00033 **17.** (a) $x + \frac{1}{3}x^3$ (b) 0.06

19. (a) $1 + x^2$ (b) 0.00006

21. (a) $8 + \frac{3}{8}(x-16) - \frac{3}{1024}(x-16)^2 + \frac{5}{65{,}536}(x-16)^3$

(b) 0.0000034 **23.** 0.57358 **25.** 3

27. $-1.037 < x < 1.037$ **29.** 21 m, no

Chapter 12 Review □ page 810

True-False Quiz

1. False **3.** False **5.** False **7.** False **9.** True
11. True **13.** False **15.** True

Exercises

1. $\frac{1}{2}$ **3.** D **5.** D **7.** e^{12} **9.** 2 **11.** C
13. C **15.** C **17.** C **19.** C **21.** C **23.** CC

25. AC **27.** 8 **29.** $\pi/4$ **31.** $\frac{4111}{3330}$ **33.** 0.9721

35. 0.18976224, error $< 6.4 \times 10^{-7}$ **39.** 4, $[-6, 2)$

41. 0.5, [2.5, 3.5)

43. $\frac{1}{2}\sum_{n=0}^{\infty}(-1)^n\left[\frac{1}{(2n)!}\left(x - \frac{\pi}{6}\right)^{2n} + \frac{\sqrt{3}}{(2n+1)!}\left(x - \frac{\pi}{6}\right)^{2n+1}\right]$

45. $\sum_{n=0}^{\infty}(-1)^n x^{n+2}$, 1 **47.** $-\sum_{n=1}^{\infty}\frac{x^n}{n}$, 1

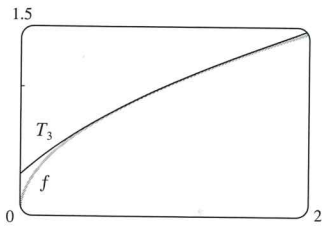
49. $\sum_{n=0}^{\infty}(-1)^n\frac{x^{8n+4}}{(2n+1)!}$, ∞

51. $\frac{1}{2} + \sum_{n=1}^{\infty}\frac{1 \cdot 5 \cdot 9 \cdot \cdots \cdot (4n-3)}{n!\,2^{6n+1}}x^n$, 16

53. $\ln|x| + C + \sum_{n=1}^{\infty}\frac{x^n}{n \cdot n!}$

55. (a) $1 + \frac{1}{2}(x-1) - \frac{1}{8}(x-1)^2 + \frac{1}{16}(x-1)^3$

(b)



(c) 0.000006

57. $-\frac{1}{6}$

Problems Plus □ page 812

1. $15!/5! = 10{,}897{,}286{,}400$

3. (b) 0 if $x = 0$, $(1/x) - \cot x$ if $x \neq n\pi$, n an integer

5. (a) $s_n = 3 \cdot 4^n$, $l_n = 1/3^n$, $p_n = 4^n/3^{n-1}$ (c) $2\sqrt{3}/5$

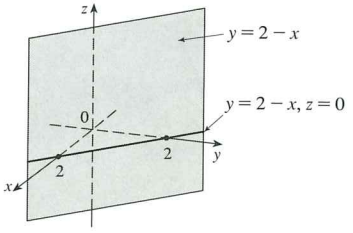
9. $(-1, 1)$, $(x^3 + 4x^2 + x)/(1-x)^4$

Chapter 13

Exercises 13.1 □ page 821

1. $(4, 0, -3)$ **3.** Q; R

5. A vertical plane that intersects the xy-plane in the line $y = 2 - x$, $z = 0$ (see graph at right)



9. (a) Yes (b) No

11. $x^2 + (y-1)^2 + (z+1)^2 = 16$; $(y-1)^2 + (z+1)^2 = 16$, $x = 0$

13. $(x-3)^2 + (y-8)^2 + (z-1)^2 = 30$ **15.** $(-1, -4, 2)$, 7

17. $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$, $\sqrt{3}/2$ **19.** (b) $\sqrt{85}/2$, $\frac{5}{2}$, $\sqrt{94}/2$

21. (a) $(x-2)^2 + (y+3)^2 + (z-6)^2 = 36$

(b) $(x-2)^2 + (y+3)^2 + (z-6)^2 = 4$

(c) $(x-2)^2 + (y+3)^2 + (z-6)^2 = 9$

23. A plane parallel to the yz-plane and 9 units in front of it

25. A half-space consisting of all points to the right of the plane $y = 2$

27. All points on and between the horizontal planes $z = 2$ and $z = -2$

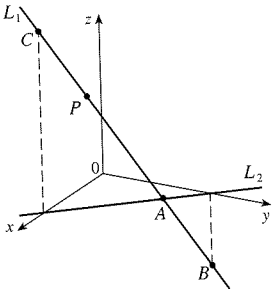
29. All points outside the sphere with radius 1 and center O

31. All points inside the sphere with radius 2 and center (0, 0, 1)

33. All points on and inside a circular cylinder of radius 3 with axis the y-axis

35. $y < 0$ **37.** $r^2 < x^2 + y^2 + z^2 < R^2$

39. (a) (2, 1, 4) (b)



41. $14x - 6y - 10z = 9$, a plane perpendicular to AB

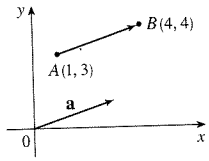
Exercises 13.2 □ page 828

1. (a) Scalar (b) Vector (c) Vector (d) Scalar

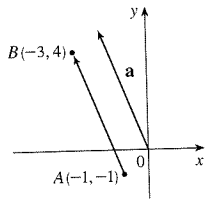
3. $\overrightarrow{AB} = \overrightarrow{DC}, \overrightarrow{DA} = \overrightarrow{CB}, \overrightarrow{DE} = \overrightarrow{EB}, \overrightarrow{EA} = \overrightarrow{CE}$

5. (a) $\overrightarrow{AC}$ (b) $\overrightarrow{CA}$ (c) $\overrightarrow{BD}$ (d) $\overrightarrow{BA}$

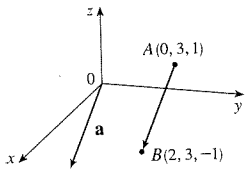
7. $\mathbf{a} = \langle 3, 1 \rangle$



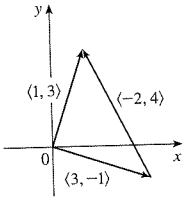
9. $\mathbf{a} = \langle -2, 5 \rangle$



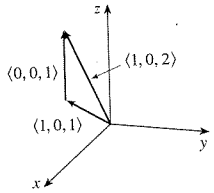
11. $\mathbf{a} = \langle 2, 0, -2 \rangle$



13. $\langle 1, 3 \rangle$



15. $\langle 1, 0, 2 \rangle$



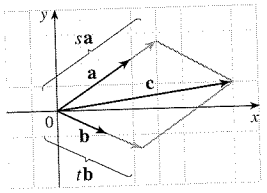
17. 5, $\langle 2, 5 \rangle$, $\langle -10, 1 \rangle$, $\langle -8, 6 \rangle$, $\langle 12, 17 \rangle$

19. 7, $\langle 5, 7, 1 \rangle$, $\langle 7, -3, 5 \rangle$, $\langle 12, 4, 6 \rangle$, $\langle 14, 26, 1 \rangle$

21. $\sqrt{6}$, $\mathbf{i} - \mathbf{j} + 3\mathbf{k}$, $\mathbf{i} - 3\mathbf{j} - \mathbf{k}$, $2\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$, $3\mathbf{i} - 2\mathbf{j} + 11\mathbf{k}$

23. $\langle 9/\sqrt{106}, -5/\sqrt{106} \rangle$ **25.** $\frac{8}{9}\mathbf{i} - \frac{1}{9}\mathbf{j} + \frac{4}{9}\mathbf{k}$ **27.** 0

29. (a), (b)



(d) $s = \frac{9}{7}, t = \frac{11}{7}$

31. $\mathbf{F} = (6\sqrt{3} - 5\sqrt{2})\mathbf{i} + (6 + 5\sqrt{2})\mathbf{j} \approx 3.32\mathbf{i} + 13.07\mathbf{j}$, $|\mathbf{F}| \approx 13.5$ lb, $\theta \approx 76°$

33. $\sqrt{493} \approx 22.2$ mi/h N8°W

35. $\mathbf{T}_1 \approx -196\,\mathbf{i} + 3.92\,\mathbf{j}$, $\mathbf{T}_2 \approx 196\,\mathbf{i} + 3.92\,\mathbf{j}$

37. A sphere with radius 1, centered at (x_0, y_0, z_0)

Exercises 13.3 □ page 836

1. (b), (c), (d) are meaningful **3.** 6 **5.** −5 **7.** 32

9. $90\sqrt{3}$ **11.** $\mathbf{u} \cdot \mathbf{v} = \frac{1}{2}$, $\mathbf{u} \cdot \mathbf{w} = -\frac{1}{2}$ **15.** $\cos^{-1}(\frac{63}{65}) \approx 14°$

17. $\cos^{-1}(1/\sqrt{238}) \approx 86°$ **19.** $\cos^{-1}(-1/(2\sqrt{7})) \approx 101°$

21. 114°, 33°, 33° **23.** Parallel **25.** Neither

27. Orthogonal **29.** 0, 6

31. $(\mathbf{i} - \mathbf{j} - \mathbf{k})/\sqrt{3}$ [or $(-\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}$]

33. $\frac{1}{3}, \frac{2}{3}, \frac{2}{3}$; 71°, 48°, 48°

35. $-8/\sqrt{77}, 3/\sqrt{77}, 2/\sqrt{77}$; 156°, 70°, 77°

37. $5/\sqrt{38}, 3/\sqrt{38}, 2/\sqrt{38}$; 36°, 61°, 71°

39. $11/\sqrt{13}$, $\langle 22/13, 33/13 \rangle$ **41.** $3/\sqrt{5}$, $\langle 6/5, 3/5, 0 \rangle$

43. $1/\sqrt{2}$, $(\mathbf{i} + \mathbf{k})/2$

47. $\langle 0, 0, -2\sqrt{10} \rangle$ or any vector of the form $\langle s, t, 3s - 2\sqrt{10} \rangle$, $s, t \in \mathbb{R}$

49. 38 J **51.** 250 cos 20° ≈ 235 ft-lb **53.** $\frac{13}{5}$

55. $\cos^{-1}(1/\sqrt{3}) \approx 55°$

Exercises 13.4 □ page 843

1. $2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ **3.** $-\mathbf{i} - \mathbf{j} + 5\mathbf{k}$ **5.** $3\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$

7. $2\mathbf{i} + 13\mathbf{j} - 8\mathbf{k}$

9. (a) Scalar (b) Meaningless (c) Vector (d) Meaningless (e) Meaningless (f) Scalar

11. 24; into the page **13.** $\langle -2, 6, -3 \rangle$, $\langle 2, -6, 3 \rangle$

15. $\langle -2/\sqrt{6}, -1/\sqrt{6}, 1/\sqrt{6} \rangle$, $\langle 2/\sqrt{6}, 1/\sqrt{6}, -1/\sqrt{6} \rangle$ **23.** 4

25. (a) $\langle 6, 3, 2 \rangle$ (b) $\frac{7}{2}$

27. (a) $\langle -10, -3, 7 \rangle$ (b) $\sqrt{158}/2$ **29.** 226 **31.** 21

35. 10.8 sin 80° ≈ 10.6 J **37.** ≈417 N **39.** (b) $\sqrt{97/3}$

45. (a) No (b) No (c) Yes

Exercises 13.5 □ page 852

1. (a) True (b) False (c) True (d) False (e) False (f) True (g) False (h) True (i) True (j) False (k) True

3. $\mathbf{r} = (-2\mathbf{i} + 4\mathbf{j} + 10\mathbf{k}) + t(3\mathbf{i} + \mathbf{j} - 8\mathbf{k})$; $x = -2 + 3t$, $y = 4 + t$, $z = 10 - 8t$

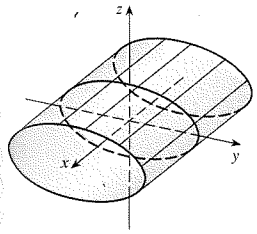
5. $\mathbf{r} = (\mathbf{i} + 6\mathbf{k}) + t(\mathbf{i} + 3\mathbf{j} + \mathbf{k})$; $x = 1 + t$, $y = 3t$, $z = 6 + t$

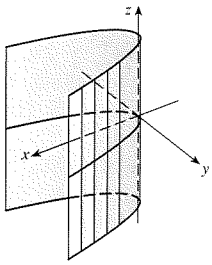
7. $x = 3, y = 1 + t, z = -1 - 5t$; $x = 3, y - 1 = (z + 1)/(-5)$

9. $x = 2 + 2t, y = 1 + \frac{1}{2}t, z = -3 - 4t$;
$(x - 2)/2 = 2y - 2 = (z + 3)/(-4)$
13. (a) $x/2 = (y - 2)/3 = (z + 1)/(-7)$
(b) $(-\frac{2}{7}, \frac{11}{7}, 0), (-\frac{4}{3}, 0, \frac{11}{3}), (0, 2, -1)$
15. Skew **17.** Parallel **19.** $-2x + y + 5z = 1$
21. $x + y - z = -1$ **23.** $2x - y + 3z = 0$
25. $3x - 7z = -9$ **27.** $x + y + z = 2$
29. $-13x + 17y + 7z = -42$ **31.** $33x + 10y + 4z = 190$
33. $x - 2y + 4z = -1$ **35.** $(1, 0, 0)$ **37.** $(-3, -1, -2)$
39. $1, 0, -1$ **41.** Neither, 60° **43.** Perpendicular
45. Parallel
47. (a) $x - 2 = y/(-8) = z/(-7)$
(b) $\cos^{-1}(-\sqrt{6}/5) \approx 119°$ (or 61°)
49. $x = 6t, y = -\frac{1}{6} + t, z = -\frac{1}{6} + 7t$ **51.** $x = z$
53. $(x/a) + (y/b) + (z/c) = 1$
55. $x = 3t, y = 1 - t, z = 2 - 2t$
57. P_1 and P_3 are parallel, P_2 and P_4 are identical **59.** $\sqrt{22/5}$
61. $\frac{25}{3}$ **63.** $7\sqrt{6}/18$ **67.** $1/\sqrt{6}$

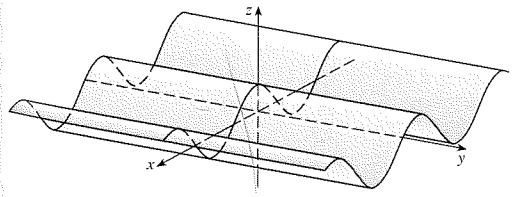
Exercises 13.6 □ page 859

1. (a) Parabola
(b) Parabolic cylinder with rulings parallel to the z-axis
(c) Parabolic cylinder with rulings parallel to the x-axis
3. Elliptic cylinder **5.** Parabolic cylinder

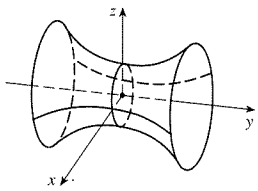




7. Cylindrical surface

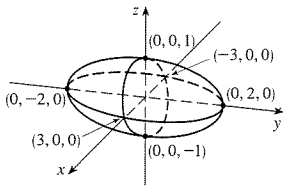


9. $x = k, z^2 - y^2 = 1 - k^2$, hyperbola
$y = k, x^2 + z^2 = 1 + k^2$, circle
$z = k, x^2 - y^2 = 1 - k^2$,
hyperbola
Hyperboloid of one sheet
with axis the y-axis

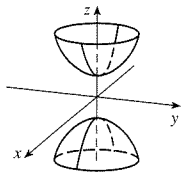


11. $x = k, y^2 + 4z^2 = 4 - 4k^2/9$, ellipse $(|k| < 3)$
$y = k, x^2 + 9z^2 = 9 - 9k^2/4$, ellipse $(|k| < 2)$
$z = k, 4x^2 + 9y^2 = 36(1 - k^2)$, ellipse $(|k| < 1)$

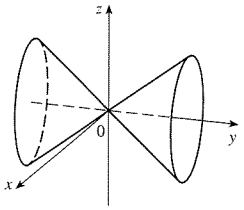
Ellipsoid



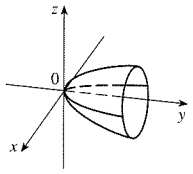
13. $x = k, 4z^2 - y^2 = 1 + k^2$, hyperbola
$y = k, 4z^2 - x^2 = 1 + k^2$, hyperbola
$z = k, x^2 + y^2 = 4k^2 - 1$, circle $(|k| > \frac{1}{2})$
Hyperboloid of two sheets



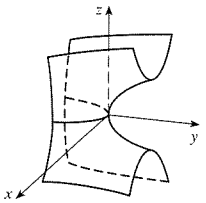
15. $x = k, y^2 - z^2 = k^2$, hyperbola $(k \neq 0)$
$y = k, x^2 + z^2 = k^2$, circle $(k \neq 0)$
$z = k, y^2 - x^2 = k^2$,
hyperbola $(k \neq 0)$
$x = 0, y = \pm z$, lines
Cone



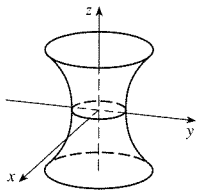
17. $x = k, y = 4z^2 + k^2$, parabola
$y = k, x^2 + 4z^2 = k$, ellipse $(k > 0)$
$z = k, y = x^2 + 4k^2$, parabola
Elliptic paraboloid with
axis the y-axis



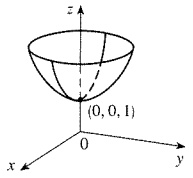
19. $x = k, y = z^2 - k^2$, parabola
$y = k, z^2 - x^2 = k$, hyperbola $(k \neq 0)$
$z = k, y = k^2 - x^2$, parabola
Hyperbolic paraboloid



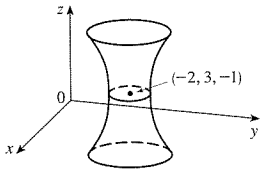
21. VII **23.** II **25.** VI **27.** VIII
29. $(x^2/4) + (y^2/3) - (z^2/12) = 1$
Hyperboloid of one sheet with
axis the z-axis



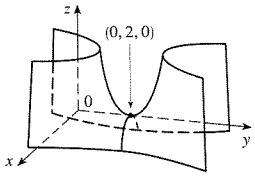
31. $z - 1 = x^2 + y^2$
Circular paraboloid with
vertex (0, 0, 1) and
axis the z-axis



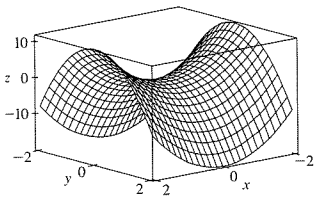
33. $(x + 2)^2 + (y - 3)^2 - 4(z + 1)^2 = 22$
Hyperboloid of one sheet
with center $(-2, 3, -1)$ and
axis parallel to the z-axis



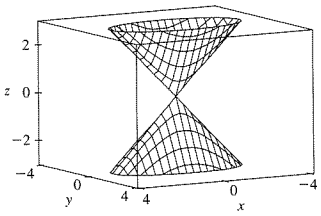
35. $z = (y - 2)^2 - x^2$
Hyperbolic paraboloid



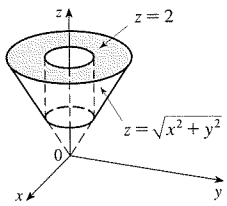
37.



39.



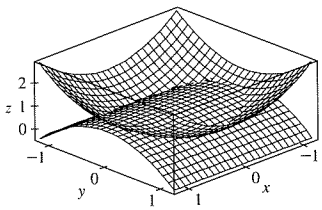
41.



43. $y = x^2 + z^2$

45. $-4x = y^2 + z^2$, paraboloid

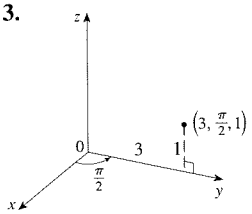
49.



Exercises 13.7 □ page 865

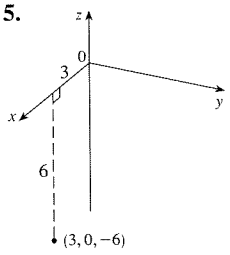
1. See pages 861–2.

3.



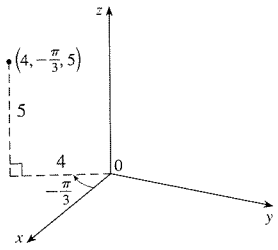
(0, 3, 1)

5.



(3, 0, −6)

7.

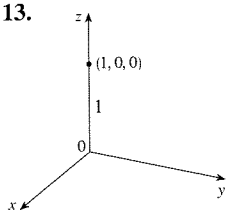


$(2, -2\sqrt{3}, 5)$

9. $(\sqrt{2}, 7\pi/4, 4)$

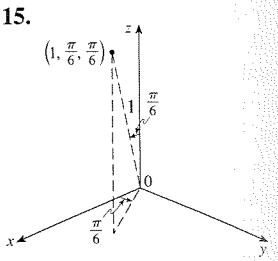
11. $(2, 4\pi/3, 2)$

13.



(0, 0, 1)

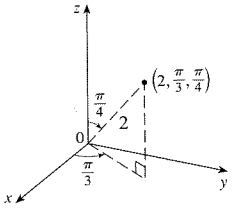
15.



$(\sqrt{3}/4, \frac{1}{4}, \sqrt{3}/2)$

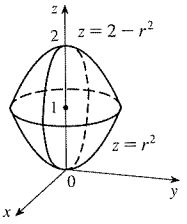
$(\frac{1}{2}\sqrt{2}, \frac{1}{2}\sqrt{6}, \sqrt{2})$

17.

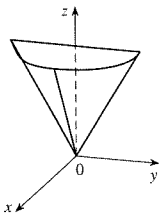


19. $(3, \pi, \pi/2)$ **21.** $(2, 0, \pi/3)$ **23.** $(\sqrt{2}, \pi/4, \pi/2)$
25. $(4\sqrt{2}, \pi/3, \pi/4)$ **27.** $(0, 0, 2)$ **29.** $(8, \pi/6, 0)$
31. Circular cylinder, radius 3, axis the z-axis
33. Positive z-axis **35.** Half-cone **37.** Circular paraboloid
39. Horizontal plane
41. Circular cylinder, radius 1, axis parallel to the z-axis
43. Sphere, radius 5, center the origin
45. Circular cylinder, radius 2, axis the y-axis
47. Cylinder, radius 1, together with the z-axis
49. (a) $r^2 + z^2 = 16$ (b) $\rho = 4$
51. (a) $r\cos\theta + 2r\sin\theta + 3z = 6$
(b) $\rho(\sin\phi\cos\theta + 2\sin\phi\sin\theta + 3\cos\phi) = 6$
53. (a) $2z^2 = r^2\cos 2\theta - 4$
(b) $\rho^2(\sin^2\phi\cos 2\theta - 2\cos^2\phi) = 4$
55. (a) $r = 2\sin\theta$ (b) $\rho\sin\phi = 2\sin\theta$

57.

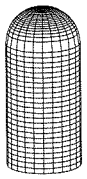


59.



61. $0 \le \phi \le \pi/4, 0 \le \rho \le \cos\phi$

63.



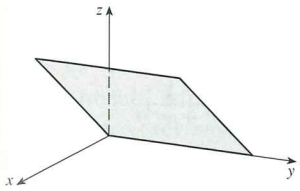
Chapter 13 Review □ page 867

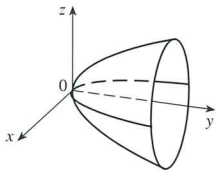
True-False Quiz

1. True **3.** True **5.** True **7.** True **9.** True
11. False **13.** False

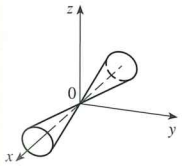
Exercises

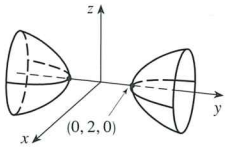
1. (a) $(x-1)^2+(y+1)^2+(z-2)^2=9$
(b) $(-2,-3,5)$, 6
3. $\mathbf{u}\cdot\mathbf{v}=3\sqrt{2}$; $|\mathbf{u}\times\mathbf{v}|=3\sqrt{2}$; out of the page **5.** $-2,-4$
7. (a) 2 (b) -2 (c) -2 (d) 0 **9.** $\cos^{-1}\left(\frac{1}{3}\right)\approx 71°$
11. (a) $\langle 4,-3,4\rangle$ (b) $\sqrt{41}/2$ **13.** 166 N, 114 N
15. $x=1+2t,\ y=2-t,\ z=4+3t$
17. $x=1+4t,\ y=-3t,\ z=1+5t$ **19.** $x+2y+5z=8$
21. (1, 4, 4) **23.** Skew **25.** $22/\sqrt{26}$
27. Plane **29.** Circular paraboloid





31. Cone **33.** Hyperboloid of two sheets





35. $4x^2+y^2+z^2=16$ **37.** $(\sqrt{3},1,2),(2\sqrt{2},\pi/6,\pi/4)$
39. $(1,\sqrt{3},2\sqrt{3}),(2,\pi/3,2\sqrt{3})$ **41.** A half-plane
43. The horizontal plane $z=3$ **45.** $r^2+z^2=4,\ \rho=2$
47. $z=4r^2$

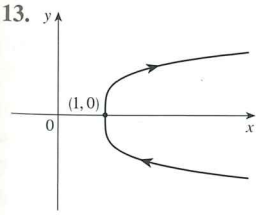
Problems Plus □ page 869

1. $(\sqrt{3}-1.5)$ m
3. (a) $(x+1)/(-2c)=(y-c)/(c^2-1)=(z-c)/(c^2+1)$
(b) $x^2+y^2=t^2+1,\ z=t$ (c) $4\pi/3$

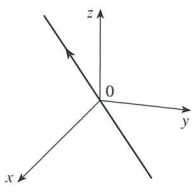
Chapter 14

Exercises 14.1 □ page 876

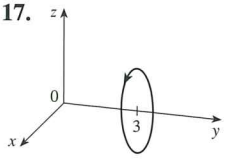
1. [1, 5] **3.** $\langle 1,0,0\rangle$ **5.** $2\mathbf{i}+\frac{1}{2}\mathbf{j}+(\tan 1)\mathbf{k}$
7. V **9.** I **11.** IV
13.



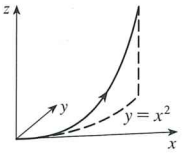
15.



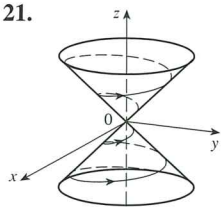
17.



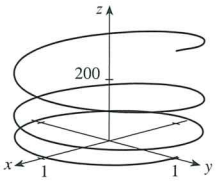
19.



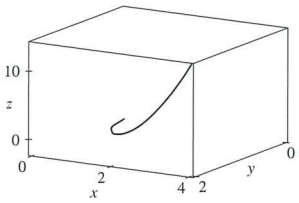
21.



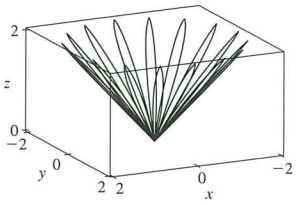
23.



25.



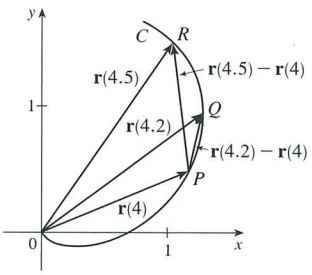
27.



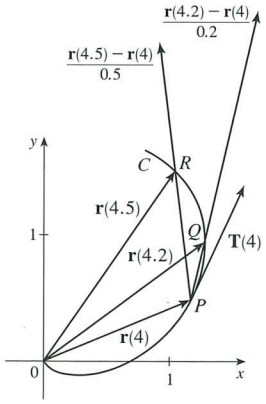
31. $\mathbf{r}(t)=t\,\mathbf{i}+\frac{1}{2}(t^2-1)\,\mathbf{j}+\frac{1}{2}(t^2+1)\,\mathbf{k},\ t\in\mathbb{R}$
33. $x=2\cos t,\ y=2\sin t,\ z=4\cos^2 t$

Exercises 14.2 □ page 882

1. (a)

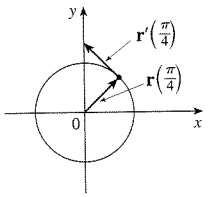


(b), (d)



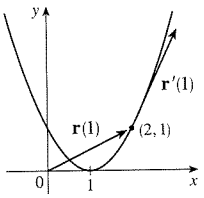
(c) $\mathbf{r}'(4)=\lim_{h\to 0}\dfrac{\mathbf{r}(4+h)-\mathbf{r}(4)}{h}$; $\mathbf{T}(4)=\dfrac{\mathbf{r}'(4)}{|\mathbf{r}'(4)|}$

3. (a), (c)

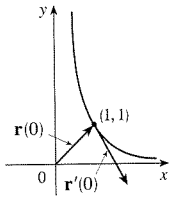


(b) $\mathbf{r}'(t) = \langle -\sin t, \cos t \rangle$

5. (a), (c)



7. (a), (c)



(b) $\mathbf{r}'(t) = \mathbf{i} + 2t\,\mathbf{j}$ (b) $\mathbf{r}'(t) = e^t\,\mathbf{i} - 2e^{-2t}\,\mathbf{j}$

9. $\mathbf{r}'(t) = \langle 2t, -1, 1/(2\sqrt{t}) \rangle$ **11.** $\mathbf{r}'(t) = 4e^{4t}\,\mathbf{k}$

13. $\mathbf{r}'(t) = -2t/(4 - t^2)\,\mathbf{i} + 1/(2\sqrt{1+t})\,\mathbf{j} - 12e^{3t}\,\mathbf{k}$

15. $\mathbf{r}'(t) = \mathbf{b} + 2t\mathbf{c}$ **17.** $\langle 15/\sqrt{262}, 6/\sqrt{262}, 1/\sqrt{262} \rangle$

19. $\frac{2}{5}\,\mathbf{i} + (2\sqrt{3}/5)\,\mathbf{j} - \frac{3}{5}\,\mathbf{k}$

21. $\langle 1, 2t, 3t^2 \rangle$, $\langle 1/\sqrt{14}, 2/\sqrt{14}, 3/\sqrt{14} \rangle$, $\langle 0, 2, 6t \rangle$, $\langle 6t^2, -6t, 2 \rangle$

23. $x = 1 + 5t, y = 1 + 4t, z = 1 + 3t$

25. $x = -(\pi/2)t, y = \frac{1}{4} + t, z = 1 + 4t$

27. $x = (\pi/4) + t, y = 1 - t, z = 1 + t$

29. (a) Not smooth (b) Smooth (c) Not smooth

31. 66° **33.** $\frac{1}{2}\,\mathbf{i} + \frac{1}{3}\,\mathbf{j} + \frac{1}{4}\,\mathbf{k}$

35. $\frac{1}{2}\,\mathbf{i} + \frac{1}{2}\,\mathbf{j} + (4 - \pi)/(4\sqrt{2})\,\mathbf{k}$

37. $e^t\,\mathbf{i} + t^2\,\mathbf{j} + (t \ln t - t)\,\mathbf{k} + \mathbf{C}$

39. $(t^3/3)\,\mathbf{i} + (t^4 + 1)\,\mathbf{j} - (t^3/3)\,\mathbf{k}$

45. $1 - 4t\cos t + 11t^2 \sin t + 3t^3 \cos t$

Exercises 14.3 □ page 889

1. $20\sqrt{29}$ **3.** $e - e^{-1}$ **5.** 9.5706

7. $\mathbf{r}(t(s)) = (1 + s/\sqrt{2})\sin[\ln(1 + s/\sqrt{2})]\,\mathbf{i} + (1 + s/\sqrt{2})\cos[\ln(1 + s/\sqrt{2})]\,\mathbf{j}$

9. $\mathbf{r}(t(s)) = 3\sin(s/5)\,\mathbf{i} + (4s/5)\,\mathbf{j} + 3\cos(s/5)\,\mathbf{k}$

11. (a) $\langle (2/\sqrt{29})\cos t, 5/\sqrt{29}, (-2/\sqrt{29})\sin t \rangle$, $\langle -\sin t, 0, -\cos t \rangle$

(b) $\frac{2}{29}$

13. (a) $\langle t^2, 2t, 2 \rangle/(t^2 + 2)$, $\langle 2t, 2 - t^2, -2t \rangle/(t^2 + 2)$

(b) $2/(t^2 + 2)^2$

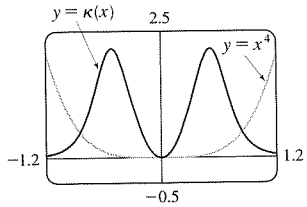
15. $2/(4t^2 + 1)^{3/2}$ **17.** $\sqrt{2}/(1 + \cos^2 t)^{3/2}$ **19.** $1/(2\sqrt{2})$

21. $6|x|/(1 + 9x^4)^{3/2}$ **23.** $|\sin x|/(1 + \cos^2 x)^{3/2}$

25. $(-\frac{1}{2}\ln 2, 1/\sqrt{2})$; approaches 0

27. (a) P (b) ≈1.3, ≈0.7

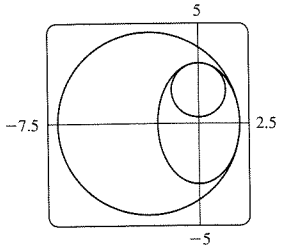
29.



31. $6/[|t|(9t^2 + 4)^{3/2}]$ **33.** $\langle \frac{2}{3}, \frac{2}{3}, \frac{1}{3} \rangle, \langle -\frac{1}{3}, \frac{2}{3}, -\frac{2}{3} \rangle, \langle -\frac{2}{3}, \frac{1}{3}, \frac{2}{3} \rangle$

35. $y = 6x + \pi, x + 6y = 6\pi$

37. $(x + \frac{5}{2})^2 + y^2 = \frac{81}{4}, x^2 + (y - \frac{5}{3})^2 = \frac{16}{9}$



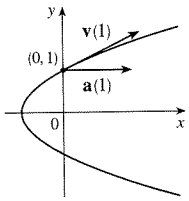
39. $(-1, -3, 1)$ **47.** $2/(t^4 + 4t^2 + 1)$ **49.** 2 m

Exercises 14.4 □ page 899

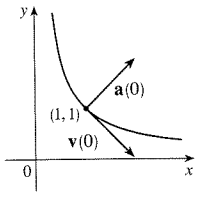
1. (a) $1.8\,\mathbf{i} - 3.8\,\mathbf{j} - 0.7\,\mathbf{k}$, $2.0\,\mathbf{i} - 2.4\,\mathbf{j} - 0.6\,\mathbf{k}$, $2.8\,\mathbf{i} + 1.8\,\mathbf{j} - 0.3\,\mathbf{k}$, $2.8\,\mathbf{i} + 0.8\,\mathbf{j} - 0.4\,\mathbf{k}$

(b) $2.4\,\mathbf{i} - 0.8\,\mathbf{j} - 0.5\,\mathbf{k}$, 2.58

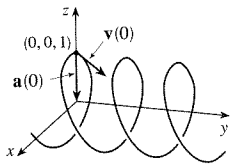
3. $\mathbf{v}(t) = \langle 2t, 1 \rangle$
$\mathbf{a}(t) = \langle 2, 0 \rangle$
$|\mathbf{v}(t)| = \sqrt{4t^2 + 1}$



5. $\mathbf{v}(t) = e^t\,\mathbf{i} - e^{-t}\,\mathbf{j}$
$\mathbf{a}(t) = e^t\,\mathbf{i} + e^{-t}\,\mathbf{j}$
$|\mathbf{v}(t)| = \sqrt{e^{2t} + e^{-2t}}$



7. $\mathbf{v}(t) = \cos t\,\mathbf{i} + \mathbf{j} - \sin t\,\mathbf{k}$
$\mathbf{a}(t) = -\sin t\,\mathbf{i} - \cos t\,\mathbf{k}$
$|\mathbf{v}(t)| = \sqrt{2}$



9. $\langle 1, 2t, 3t^2 \rangle$, $\langle 0, 2, 6t \rangle$, $\sqrt{1 + 4t^2 + 9t^4}$

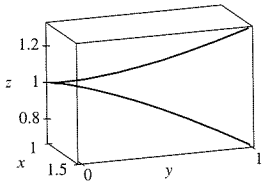
11. $\sqrt{2}\,\mathbf{i} + e^t\,\mathbf{j} - e^{-t}\,\mathbf{k}$, $e^t\,\mathbf{j} + e^{-t}\,\mathbf{k}$, $e^t + e^{-t}$

13. $e^t[(\cos t - \sin t)\,\mathbf{i} + (\sin t + \cos t)\,\mathbf{j} + (t + 1)\,\mathbf{k}]$, $e^t[-2\sin t\,\mathbf{i} + 2\cos t\,\mathbf{j} + (t + 2)\,\mathbf{k}]$, $e^t\sqrt{t^2 + 2t + 3}$

15. $\mathbf{v}(t) = \mathbf{i} - \mathbf{j} + t\,\mathbf{k}$, $\mathbf{r}(t) = t\,\mathbf{i} - t\,\mathbf{j} + (t^2/2)\,\mathbf{k}$

17. (a) $\mathbf{r}(t) = (1 + t^2/2)\,\mathbf{i} + t^2\,\mathbf{j} + (1 + t^3/3)\,\mathbf{k}$

(b)



19. $t = 4$ **21.** $\mathbf{r}(t) = t\,\mathbf{i} - t\,\mathbf{j} + \frac{5}{2}t^2\,\mathbf{k}$, $|\mathbf{v}(t)| = \sqrt{25t^2 + 2}$

23. (a) 22 km (b) 3.2 km (c) 500 m/s **25.** 30 m/s

29. $6t, 6$ **31.** $0, 1$ **33.** $e^t - e^{-t}, \sqrt{2}$ **35.** $t = 1$

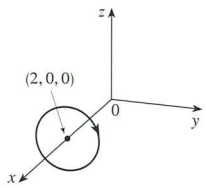
Chapter 14 Review □ page 902

True-False Quiz

1. True **3.** False **5.** False **7.** False **9.** True

Exercises

1. (a)



(b) $\cos t\,\mathbf{j} - \sin t\,\mathbf{k}$, $-\sin t\,\mathbf{j} - \cos t\,\mathbf{k}$

3. $\mathbf{r}(t) = 4\cos t\,\mathbf{i} + 4\sin t\,\mathbf{j} + (5 - 4\cos t)\,\mathbf{k}$, $0 \leqslant t \leqslant 2\pi$

5. $\frac{5}{6}\mathbf{i} + \frac{9}{4}\mathbf{j} + \frac{1}{5}\mathbf{k}$ 7. 15.9241 9. $\pi/2$

11. (a) $\langle t^2, t, 1\rangle/\sqrt{t^4 + t^2 + 1}$

(b) $\langle 2t, 1 - t^4, -2t^3 - t\rangle/\sqrt{t^8 + 4t^6 + 2t^4 + 5t^2}$

(c) $\sqrt{t^8 + 4t^6 + 2t^4 + 5t^2}/(t^4 + t^2 + 1)^2$

13. $12/17^{3/2}$ 15. $x - 2y + 2\pi = 0$

17. $\mathbf{v}(t) = 2\sqrt{2}\,\mathbf{i} + 2e^{2t}\,\mathbf{j} - 2e^{-2t}\,\mathbf{k}$, $|\mathbf{v}(t)| = 2(e^{2t} + e^{-2t})$, $\mathbf{a}(t) = 4e^{2t}\,\mathbf{j} + 4e^{-2t}\,\mathbf{k}$

19. (a) About 3.8 ft above the ground, 60.8 ft from the athlete

(b) ≈21.4 ft (c) ≈64.2 ft from the athlete

21. (c) $-2e^{-t}\,\mathbf{v}_d + e^{-t}\,\mathbf{R}$

Problems Plus □ page 904

1. (a) $\mathbf{v} = \omega R(-\sin \omega t\,\mathbf{i} + \cos \omega t\,\mathbf{j})$ (c) $\mathbf{a} = -\omega^2\mathbf{r}$

3. (a) 90°, $v_0^2/(2g)$

5. (b) $\mathbf{R}(t) = (m/k)(1 - e^{-kt/m})\mathbf{v}_0 + (gm/k)[(m/k)(1 - e^{-kt/m}) - t]\mathbf{j}$

7. (a) ≈0.94 ft to the right of the table's edge, ≈15 ft/s

(b) ≈7.6° (c) ≈2.13 ft to the right of the table's edge

Chapter 15

Exercises 15.1 □ page 917

1. (a) −7; a temperature of 8 °C with wind blowing at 60 km/h feels equivalent to about −7 °C without wind.

(b) When the temperature is −12 °C, what wind speed gives a wind-chill of −26 °C? 20 km/h

(c) With a wind speed of 80 km/h, what temperature gives a wind-chill of −14 °C? 4 °C

(d) A function of wind speed that gives wind-chill values when the temperature is −4 °C

(e) A function of temperature that gives wind-chill values when the wind speed is 50 km/h

3. Yes

5. (a) 25; a 40-knot wind blowing in the open sea for 15 h will create waves about 25 ft high.

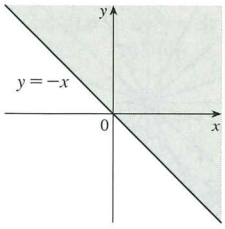
(b) $f(30, t)$ is a function of t giving the wave heights produced by 30-knot winds blowing for t hours.

(c) $f(v, 30)$ is a function of v giving the wave heights produced by winds of speed v blowing for 30 hours.

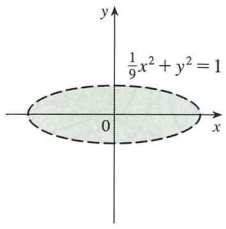
7. (a) 1 (b) $\mathbb{R}^2$ (c) $\{z \mid z > 0\}$

9. (a) 0 (b) $\{(x, y, z) \mid x - y + z > 0\}$ (c) $\mathbb{R}$

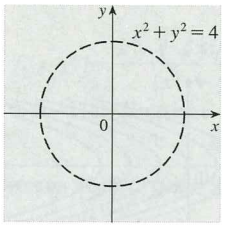
11. $\{(x, y) \mid y \geqslant -x\}$



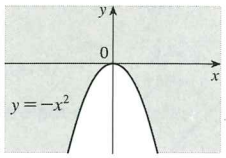
13. $\{(x, y) \mid \frac{1}{9}x^2 + y^2 < 1\}$



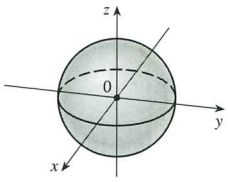
15. $\{(x, y) \mid x^2 + y^2 \neq 4\}$



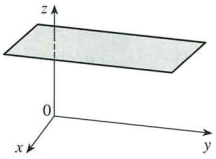
17. $\{(x, y) \mid y \geqslant -x^2\}$



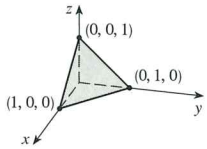
19. $\{(x, y, z) \mid x^2 + y^2 + z^2 \leqslant 1\}$



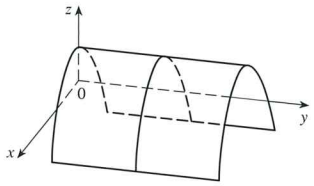
21. $z = 3$, horizontal plane



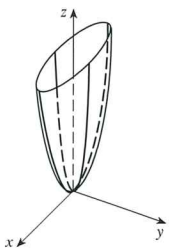
23. $x + y + z = 1$, plane



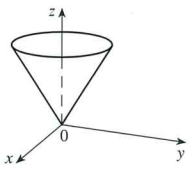
25. $z = 1 - x^2$



27. $z = x^2 + 9y^2$, elliptic paraboloid

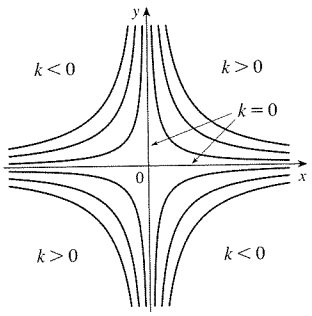


29. $z = \sqrt{x^2 + y^2}$, top half of cone

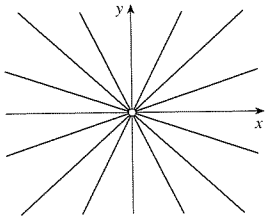


31. ≈56, ≈35 33. Steep; nearly flat

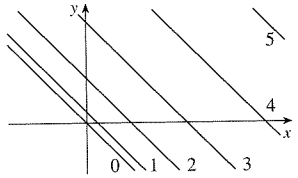
35. $xy = k$



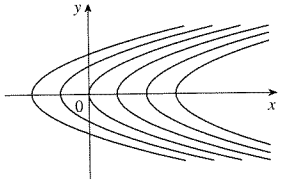
37. $x/y = k$



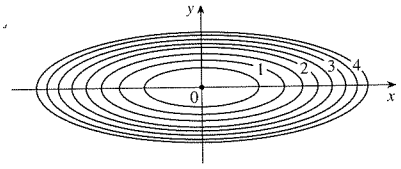
39. $\sqrt{x + y} = k$

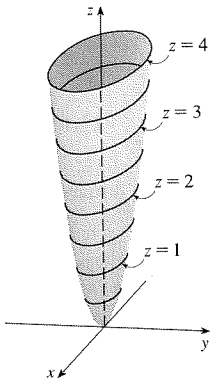


41. $x = y^2 + k$

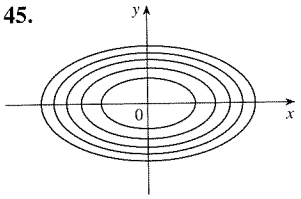


43. $x^2 + 9y^2 = k$

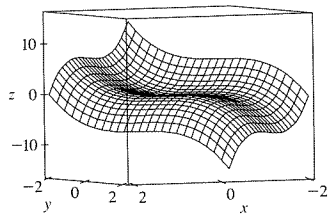




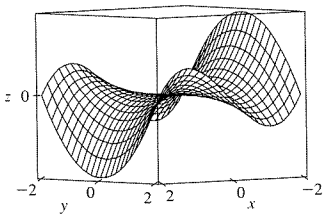
45.



47.

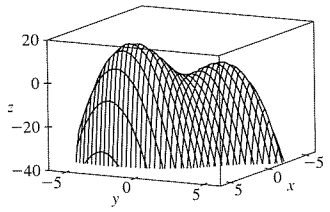


49.



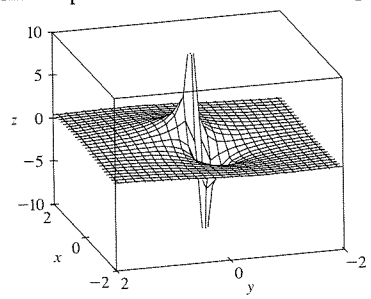
51. (a) B (b) III **53.** (a) F (b) V
55. (a) D (b) IV **57.** Family of parallel planes
59. Family of hyperboloids of one or two sheets with axis the y-axis

61.



f appears to have a maximum value of about 15. There are 2 local maximum points but no local minimum point.

63.



The function values approach 0 as x, y become large; as (x, y) approaches the origin, f approaches $\pm\infty$ or 0, depending on the direction of approach.

65. If $c = 0$, the graph is a cylindrical surface. For $c > 0$, the level curves are ellipses. The graph curves upward as we leave the origin, and the steepness increases as c increases. For $c < 0$, the level curves are hyperbolas. The graph curves upward in the y-direction and downward, approaching the xy-plane, in the x-direction giving a saddle-shaped appearance near $(0, 0, 1)$.

67. (b) $y = 0.75x + 0.01$

Exercises 15.2 □ page 928

1. Nothing; if f is continuous, $f(3, 1) = 6$ **3.** $-\frac{5}{2}$ **5.** 2025
7. Does not exist **9.** Does not exist **11.** 0
13. Does not exist **15.** 2 **17.** 1 **19.** Does not exist
21. The graph shows that the function approaches different numbers along different lines.
23. $h(x, y) = 4x^2 + 9y^2 + 12xy - 24x - 36y + 36 + \sqrt{2x + 3y - 6}$; $\{(x, y) \mid 2x + 3y \geqslant 6\}$
25. Along the line $y = x$ **27.** $\{(x, y) \mid y \neq x^2\}$
29. $\{(x, y) \mid y \geqslant 0\}$ **31.** $\{(x, y) \mid x \geqslant |y|\}$
33. $\{(x, y, z) \mid z \neq x^2 + y^2\}$ **35.** $\{(x, y) \mid (x, y) \neq (0, 0)\}$
37. 0 **39.** 0

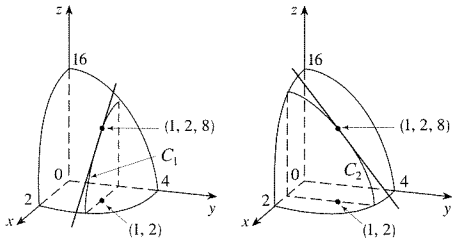
Exercises 15.3 □ page 939

1. (a) The rate of change of temperature as longitude varies, with latitude and time fixed; the rate of change as only latitude varies; the rate of change as only time varies.
(b) Positive, negative, positive
3. (a) $f_T(12, 20) \approx 1.375$; for a temperature of 12 °C and wind speed of 20 km/h, the wind-chill index rises by 1.375 °C for each degree the temperature increases.
$f_v(12, 20) \approx -0.3$; for a temperature of 12 °C and wind speed of 20 km/h, the wind-chill index decreases by 0.3 °C for each km/h the wind speed increases.

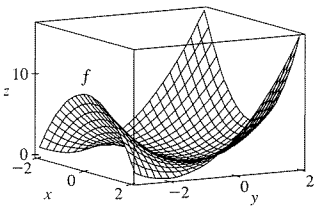
(b) Positive, negative (c) 0

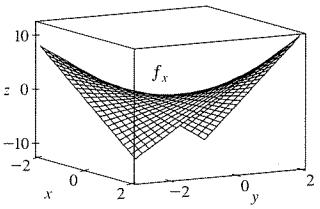
5. $c = f, b = f_x, a = f_y$

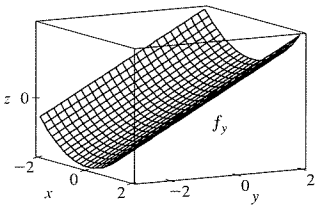
7. $f_x(1, 2) = -8$ = slope of C_1, $f_y(1, 2) = -4$ = slope of C_2



9. $f_x = 2x + 2xy, f_y = 2y + x^2$







11. $f_x(x, y) = 3, f_y(x, y) = -8y^3$

13. $\partial z/\partial x = e^{3y}, \partial z/\partial y = 3xe^{3y}$

15. $f_x(x, y) = 2y/(x + y)^2, f_y(x, y) = -2x/(x + y)^2$

17. $\partial w/\partial \alpha = \cos \alpha \cos \beta, \partial w/\partial \beta = -\sin \alpha \sin \beta$

19. $f_u = v/(u^2 + v^2), f_v = -u/(u^2 + v^2)$

21. $\partial z/\partial x = 1/\sqrt{x^2 + y^2}, \partial z/\partial y = y/\left(x^2 + y^2 + x\sqrt{x^2 + y^2}\right)$

23. $f_x = y^2z^3, f_y = 2xyz^3 + 3z, f_z = 3xy^2z^2 + 3y$

25. $\partial w/\partial x = 1/(x + 2y + 3z), \partial w/\partial y = 2/(x + 2y + 3z)$, $\partial w/\partial z = 3/(x + 2y + 3z)$

27. $\partial u/\partial x = e^{-t} \sin \theta, \partial u/\partial t = -xe^{-t} \sin \theta, \partial u/\partial \theta = xe^{-t} \cos \theta$

29. $f_x = 1/(z - t), f_y = 1/(t - z), f_z = (y - x)/(z - t)^2$, $f_t = (x - y)/(z - t)^2$

31. $\partial u/\partial x_i = x_i/\sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$ **33.** $\frac{3}{5}$ **35.** $-\frac{1}{3}$

37. $f_x(x, y) = 2x - y, f_y(x, y) = 4y - x$

39. $(y - z)/(x - y), (x + z)/(x - y)$

41. $(x - y - z)/(x + z), (y - x)/(x + z)$

43. (a) $f'(x), g'(y)$ (b) $f'(x + y), f'(x + y)$

45. $f_{xx} = 12x^2 - 6y^3, f_{xy} = -18xy^2 = f_{yx}, f_{yy} = -18x^2y$

47. $z_{xx} = -2y/(x + y)^3, z_{xy} = (x - y)/(x + y)^3 = z_{yx}$, $z_{yy} = 2x/(x + y)^3$

49. $u_{ss} = e^{-s} \sin t, u_{st} = -e^{-s} \cos t = u_{ts}, u_{tt} = -e^{-s} \sin t$

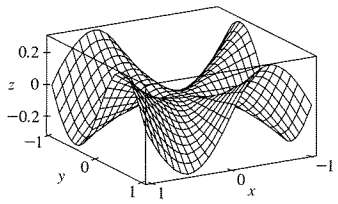
55. $-48xy$ **57.** $48x^3y^3z^2$ **59.** $-\sin y$

61. $72yz^2/(x + 2y^2 + 3z^3)^3$ **63.** $\approx 12.2, \approx 16.8, \approx 23.25$

75. R^2/R_1^2 **79.** No **81.** $x = 1 + t, y = 2, z = 2 - 2t$

85. -2

87. (a)



(b) $f_x(x, y) = \dfrac{x^4y + 4x^2y^3 - y^5}{(x^2 + y^2)^2}, f_y(x, y) = \dfrac{x^5 - 4x^3y^2 - xy^4}{(x^2 + y^2)^2}$

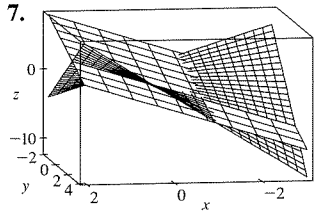
(c) 0, 0 (e) No, since f_{xy} and f_{yx} are not continuous.

Exercises 15.4 □ page 950

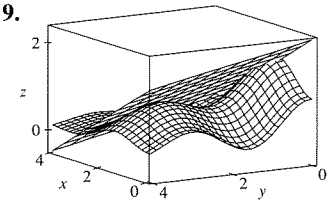
1. $8x + 10y - z = 9$ **3.** $x - 2y + z = 4$

5. $2x + y - z = 1$

7.



9.



11. $2x + \frac{1}{4}y - 1$ **13.** $x + 1$ **15.** $\frac{1}{2}x + y + \frac{1}{4}\pi - \frac{1}{2}$

17. $-\frac{2}{3}x - \frac{7}{3}y + \frac{20}{3}; 2.84\overline{6}$ **19.** $\frac{3}{7}x + \frac{2}{7}y + \frac{6}{7}z; 6.9914$

21. $4T + H - 329; 129\ ^\circ\text{F}$ **23.** $dz = 2xy^3\,dx + 3x^2y^2\,dy$

25. $du = e^t \sin \theta\,dt + e^t \cos \theta\,d\theta$

27. $dw = (x^2 + y^2 + z^2)^{-1}(x\,dx + y\,dy + z\,dz)$

29. $\Delta z = 0.9225, dz = 0.9$ **31.** $5.4\ \text{cm}^2$ **33.** $16\ \text{cm}^3$

35. 150 **37.** $\frac{1}{17} \approx 0.059\ \Omega$ **39.** $\varepsilon_1 = \Delta x, \varepsilon_2 = \Delta y$

Exercises 15.5 □ page 958

1. $4(2xy + y^2)t^3 - 3(x^2 + 2xy)t^2$

3. $\pi \cos x \cos y - (\sin x \sin y)/(2\sqrt{t})$

5. $e^{y/z}[2t - (x/z) - (2xy/z^2)]$

7. $\partial z/\partial s = 2x + y + xt + 2yt, \partial z/\partial t = 2x + y + xs + 2ys$

9. $\dfrac{\partial z}{\partial s} = \dfrac{4st + \ln t}{1 + (2x + y)^2}, \dfrac{\partial z}{\partial t} = \dfrac{2s^2 + s/t}{1 + (2x + y)^2}$

11. $\dfrac{\partial z}{\partial s} = e^r\left(t \cos \theta - \dfrac{s}{\sqrt{s^2 + t^2}} \sin \theta\right)$, $\dfrac{\partial z}{\partial t} = e^r\left(s \cos \theta - \dfrac{t}{\sqrt{s^2 + t^2}} \sin \theta\right)$

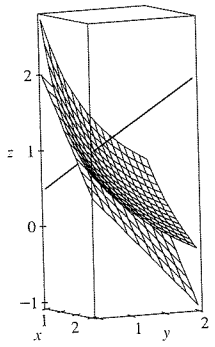
13. 62

15. $\dfrac{\partial u}{\partial r} = \dfrac{\partial u}{\partial x}\dfrac{\partial x}{\partial r} + \dfrac{\partial u}{\partial y}\dfrac{\partial y}{\partial r}, \dfrac{\partial u}{\partial s} = \dfrac{\partial u}{\partial x}\dfrac{\partial x}{\partial s} + \dfrac{\partial u}{\partial y}\dfrac{\partial y}{\partial s}$, $\dfrac{\partial u}{\partial t} = \dfrac{\partial u}{\partial x}\dfrac{\partial x}{\partial t} + \dfrac{\partial u}{\partial y}\dfrac{\partial y}{\partial t}$

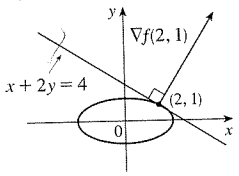
17. $\dfrac{\partial v}{\partial x} = \dfrac{\partial v}{\partial p}\dfrac{\partial p}{\partial x} + \dfrac{\partial v}{\partial q}\dfrac{\partial q}{\partial x} + \dfrac{\partial v}{\partial r}\dfrac{\partial r}{\partial x}$,
$\dfrac{\partial v}{\partial y} = \dfrac{\partial v}{\partial p}\dfrac{\partial p}{\partial y} + \dfrac{\partial v}{\partial q}\dfrac{\partial q}{\partial y} + \dfrac{\partial v}{\partial r}\dfrac{\partial r}{\partial y}$,
$\dfrac{\partial v}{\partial z} = \dfrac{\partial v}{\partial p}\dfrac{\partial p}{\partial z} + \dfrac{\partial v}{\partial q}\dfrac{\partial q}{\partial z} + \dfrac{\partial v}{\partial r}\dfrac{\partial r}{\partial z}$
19. 2, 0 **21.** 0, 0, 4
23. $\partial u/\partial p = 2(z - x)/(y + z)^2 = -t/p^2$, $\partial u/\partial r = 0$, $\partial u/\partial t = 2/(y + z) = 1/p$
25. $(y - 2x)/(3y^2 - x)$ **27.** $\dfrac{\sin(x - y) + e^y}{\sin(x - y) - xe^y}$
29. $-\dfrac{y^2 + 2xz}{2yz + x^2}, -\dfrac{2xy + z^2}{2yz + x^2}$
31. $-(e^y + ze^x)/(y + e^x)$, $-(xe^y + z)/(y + e^x)$ **33.** 2 °C/s
35. ≈ −0.33 m/s per minute
37. (a) 6 m³/s (b) 10 m²/s (c) 0 m/s **39.** −0.27 L/s
41. (a) $\partial z/\partial r = (\partial z/\partial x)\cos\theta + (\partial z/\partial y)\sin\theta$, $\partial z/\partial\theta = -(\partial z/\partial x)r\sin\theta + (\partial z/\partial y)r\cos\theta$
47. $4rs\,\partial^2 z/\partial x^2 + (4r^2 + 4s^2)\partial^2 z/\partial x\,\partial y + 4rs\,\partial^2 z/\partial y^2 + 2\,\partial z/\partial y$

Exercises 15.6 □ page 970

1. ≈ −0.1 millibar/mi **3.** $7\sqrt{3} - 16$ **5.** $\frac{5}{16}\sqrt{3} + \frac{1}{4}$
7. (a) $\nabla f(x, y) = \langle 5y^2 - 12x^2y, 10xy - 4x^3\rangle$
(b) $\langle -4, 16\rangle$ (c) 172/13
9. (a) $\nabla f(x, y, z) = \langle y^2z^3, 2xyz^3, 3xy^2z^2\rangle$ (b) $\langle 4, -4, 12\rangle$
(c) $20/\sqrt{3}$
11. 23/10 **13.** $4\sqrt{2}$ **15.** 4/9 **17.** $-\pi/(4\sqrt{3})$
19. 2/5 **21.** $\sqrt{5}$, $\langle 1, 2\rangle$ **23.** 1, $\langle 0, 1\rangle$
25. $\sqrt{11}$, $\langle 1, -1, -3\rangle$ **27.** (b) $\langle -12, 92\rangle$
29. (a) $-40/(3\sqrt{3})$
31. (a) $32/\sqrt{3}$ (b) $\langle 38, 6, 12\rangle$ (c) $2\sqrt{406}$ **33.** $\frac{327}{13}$
37. (a) $4x - 2y + 3z = 21$ (b) $\dfrac{x - 4}{8} = \dfrac{y + 1}{-4} = \dfrac{z - 1}{6}$
39. (a) $3x - y + z = 4$ (b) $(x - 1)/3 = -y = z - 1$
41. (a) $x + y - z = 1$ (b) $x - 1 = y = -z$
43.



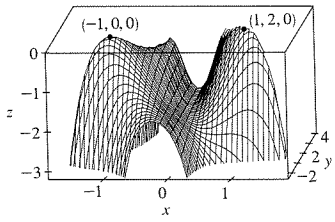
45. $\langle 4, 8\rangle$, $x + 2y = 4$



51. $(\pm\sqrt{6}/3, \mp 2\sqrt{6}/3, \pm\sqrt{6}/2)$
57. $x = -1 - 10t$, $y = 1 - 16t$, $z = 2 - 12t$
61. If $\mathbf{u} = \langle a, b\rangle$ and $\mathbf{v} = \langle c, d\rangle$, then $af_x + bf_y$ and $cf_x + df_y$ are known, so we solve linear equations for f_x and f_y.

Exercises 15.7 □ page 981

1. (a) f has a local minimum at (1, 1).
(b) f has a saddle point at (1, 1).
3. Local minimum at (1, 1), saddle point at (0, 0)
5. Maximum $f(-1, \frac{1}{2}) = 11$
7. Minimum $f(0, 0) = 4$, saddle points $(\pm\sqrt{2}, -1)$
9. f has a local maximum value of 1 at all points of the form (x_0, x_0).
11. Saddle point (1, 2) **13.** Maximum $f(-\frac{1}{2}, 4) = -6$
15. None **17.** Saddle points $(0, n\pi)$, n an integer
19. Maximum $f(0, 0) = 2$, minimum $f(0, 2) = -2$, saddle points $(\pm 1, 1)$
21. Maximum $f(\pi/3, \pi/3) = 3\sqrt{3}/2$, minimum $f(5\pi/3, 5\pi/3) = -3\sqrt{3}/2$
23. Minima $f(-1.714, 0) \approx -9.200$, $f(1.402, 0) \approx 0.242$, saddle point (0.312, 0), lowest point (−1.714, 0, −9.200)
25. Maxima $f(-1.267, 0) \approx 1.310$, $f(1.629, \pm 1.063) \approx 8.105$, saddle points (−0.259, 0), (1.526, 0), highest points (1.629, ±1.063, 8.105)
27. Minimum $f(4, 0) = -7$, maximum $f(4, 5) = 13$
29. Maximum $f(\pm 1, 1) = 7$, minimum $f(0, 0) = 4$
31. Maximum $f(2, 4) = 3$, minimum $f(-2, 4) = -9$
33. Maximum $f(1, 0) = 2$, minimum $f(-1, 0) = -2$
35.



37. $7/\sqrt{61}$ **39.** (0, 0, 1), (0, 0, −1) **41.** $\frac{100}{3}, \frac{100}{3}, \frac{100}{3}$
43. $16/\sqrt{3}$ **45.** $\frac{4}{3}$ **47.** Cube, edge length $c/12$
49. Square base of side 40 cm, height 20 cm

Exercises 15.8 □ page 990

1. ≈59, 30 **3.** Maxima $f(\pm 1, 0) = 1$, minima $f(0, \pm 1) = -1$
5. Maxima $f(\pm 2, 1) = 4$, minima $f(\pm 2, -1) = -4$
7. Maximum $f(1, 3, 5) = 70$, minimum $f(-1, -3, -5) = -70$
9. Maximum $2/\sqrt{3}$, minimum $-2/\sqrt{3}$
11. Maximum $\sqrt{3}$, minimum 1
13. Maximum $f(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}) = 2$, minimum $f(-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}) = -2$
15. Maximum $f(1, \sqrt{2}, -\sqrt{2}) = 1 + 2\sqrt{2}$, minimum $f(1, -\sqrt{2}, \sqrt{2}) = 1 - 2\sqrt{2}$
17. Maximum $\frac{3}{2}$, minimum $\frac{1}{2}$
19. Maxima $f(\pm 1/\sqrt{2}, \mp 1/(2\sqrt{2})) = e^{1/4}$, minima $f(\pm 1/\sqrt{2}, \pm 1/(2\sqrt{2})) = e^{-1/4}$
25–37. See Exercises 37–49 in Section 15.7.
39. Nearest $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, farthest $(-1, -1, 2)$
41. (a) c/n (b) When $x_1 = x_2 = \cdots = x_n$

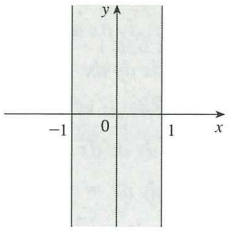
Chapter 15 Review □ page 995

True-False Quiz

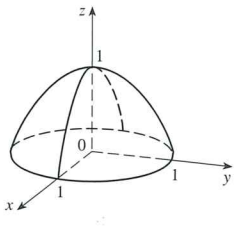
1. True **3.** False **5.** False **7.** True **9.** False **11.** True

Exercises

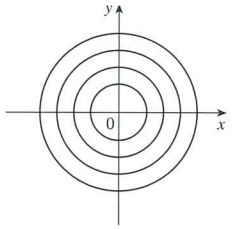
1. $\{(x, y) \mid -1 \leqslant x \leqslant 1\}$



3.



5.



7. $\frac{2}{3}$

9. (a) ≈ 3.5 °C/m, -3.0 °C/m
(b) ≈ 0.35 °C/m by Equation 15.6.9 (Definition 15.6.2 gives ≈ 1.1 °C/m.)
(c) -0.25

11. $f_x = 12x^3 - \sqrt{y}, f_y = -\frac{1}{2}x/\sqrt{y}$

13. $f_s = 2e^{2s}\cos \pi t, f_t = -\pi e^{2s}\sin \pi t$

15. $f_x = y^z, f_y = xzy^{z-1}, f_z = xy^z \ln y$

17. $f_{xx} = 2y^3 - 24x^2, f_{xy} = 6xy^2, f_{yy} = 6x^2y + 2$

19. $f_{xx} = 0, f_{yy} = 2xz^3, f_{zz} = 6xy^2z, f_{xy} = 2yz^3, f_{xz} = 3y^2z^2, f_{yz} = 6xyz^2$

23. $6y - z = 1$ **25.** $x + 3y + 9z = 18$

27. $3x + 4y + 3z = 14$ **29.** $(\pm\sqrt{2/7}, \pm 1/\sqrt{14}, \mp 3/\sqrt{14})$

31. $60x + \frac{24}{5}y + \frac{32}{5}z - 120$; 38.656

33. $e^t + 2(y/z)(3t^2 + 4) - 2t(y^2/z^2)$ **35.** $-47, 108$

41. $ze^{x\sqrt{y}}\langle z\sqrt{y}, xz/(2\sqrt{y}), 2\rangle$ **43.** $\frac{43}{5}$ **45.** $\sqrt{145}/2, \langle 4, \frac{9}{2}\rangle$

47. $\approx \frac{5}{8}$ knot/mi **49.** Minimum $f(-4, 1) = -9$

51. Maximum $f(1, 1) = 1$; saddle points $(0, 0), (0, 3), (3, 0)$

53. Maximum $f(1, 2) = 4$, minimum $f(2, 4) = -64$

55. Maximum $f(-1, 0) = 2$, minima $f(1, \pm 1) = -3$, saddle points $(-1, \pm 1), (1, 0)$

57. Maximum $f(\pm\sqrt{2/3}, 1/\sqrt{3}) = 2/(3\sqrt{3})$, minimum $f(\pm\sqrt{2/3}, -1/\sqrt{3}) = -2/(3\sqrt{3})$

59. Maximum $f(3, 3, 3) = 9$, minimum $f(1, 1, -1) = f(1, -1, 1) = f(-1, 1, 1) = 1$

61. $(\pm 3^{-1/4}, 3^{-1/4}\sqrt{2}, \pm 3^{1/4}), (\pm 3^{-1/4}, -3^{-1/4}\sqrt{2}, \pm 3^{1/4})$

63. $P(2 - \sqrt{3}), P(3 - \sqrt{3})/6, P(2\sqrt{3} - 3)/3$

Problems Plus □ page 998

1. $L^2W^2, \frac{1}{4}L^2W^2$ **3.** (a) $x = w/3$, base $= w/3$ (b) Yes

9. $\sqrt{6}/2, 3\sqrt{2}/2$

Chapter 16

Exercises 16.1 □ page 1008

1. (a) -17.75 (b) -15.75 (c) -8.75 (d) -6.75

3. (a) 63 (b) 43.5 **5.** (a) -6 (b) -3.5

7. $U < V < L$ **9.** ≈ 248 **11.** 60 **13.** 3

15. 0.6065, 0.5694, 0.5606, 0.5585, 0.5579, 0.5578

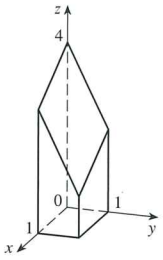
Exercises 16.2 □ page 1014

1. $9 + 27y, 8x + 24x^2$ **3.** 10 **5.** 1 **7.** $\frac{4}{15}(31 - 9\sqrt{3})$

9. $\frac{21}{2}\ln 2$ **11.** 6 **13.** $\frac{21}{2}$ **15.** $9 \ln 2$

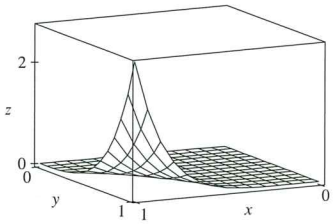
17. $[(\sqrt{3} - 1)/2] - (\pi/12)$ **19.** $\ln \frac{27}{16}$

21.



23. 37.5 **25.** $\frac{166}{27}$ **27.** $\frac{4}{15}(2\sqrt{2} - 1)$ **29.** 36

31. $21e - 57$ **33.** $\frac{5}{6}$



35. Fubini's Theorem does not apply. The integrand has an infinite discontinuity at the origin.

Exercises 16.3 □ page 1022

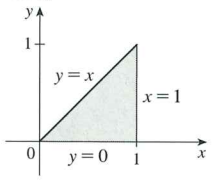
1. $\frac{9}{20}$ **3.** $\frac{4}{9}e^{3/2} - \frac{32}{45}$ **5.** $e - 1$ **7.** $\frac{256}{21}$ **9.** $\frac{1}{2}\ln 2$

11. $\frac{1}{2}e^4 - 2e$ **13.** $(1 - \cos 1)/2$ **15.** $\frac{147}{20}$ **17.** 0

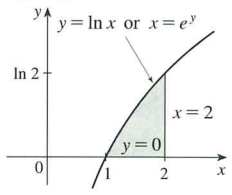
19. $\frac{6}{35}$ **21.** $\frac{31}{8}$ **23.** $\frac{1}{6}(11\sqrt{5} - 27) + \frac{9}{2}\sin^{-1}\frac{2}{3}$ **25.** $\frac{1}{6}$

27. $\frac{1}{3}$ **29.** 0, 1.213, 0.713 **31.** 13,984,735,616/14,549,535

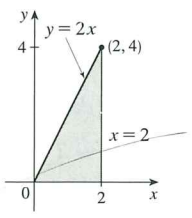
33. $\int_0^1 \int_y^1 f(x, y)\, dx\, dy$



35. $\int_0^{\ln 2} \int_{e^y}^2 f(x, y)\, dx\, dy$



37. $\int_0^2 \int_0^{2x} f(x, y)\, dy\, dx$



39. $(e^9 - 1)/6$ **41.** $\frac{1}{4}\sin 81$ **43.** $(2\sqrt{2} - 1)/3$ **45.** 1
47. $0 \le \iint_D \sqrt{x^3 + y^3}\, dA \le \sqrt{2}$ **51.** 8π **53.** $2\pi/3$

Exercises 16.4 □ page 1028

1. $\int_0^{2\pi}\int_0^2 f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta$ **3.** $\int_{-2}^2\int_x^2 f(x, y)\, dy\, dx$
5. $\int_0^{2\pi}\int_2^5 f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta$ **7.** 0 **9.** $\frac{609}{8}$
11. $(\pi/2)(1 - e^{-4})$ **13.** $24\pi^5$ **15.** $\pi/12$ **17.** 4
19. $81\pi/2$ **21.** $\frac{4}{3}\pi a^3$ **23.** $(2\pi/3)[1 - (1/\sqrt{2})]$
25. $(8\pi/3)(64 - 24\sqrt{3})$ **27.** $(\pi/4)(e - 1)$ **29.** $4\pi/3$
31. 1800π ft^3 **33.** $\frac{15}{16}$ **35.** (a) $\sqrt{\pi}/4$ (b) $\sqrt{\pi}/2$

Exercises 16.5 □ page 1038

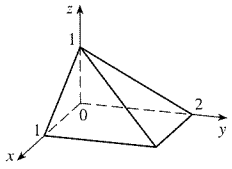
1. $\frac{50}{3}$ C **3.** $\frac{2}{3}, (0, \frac{1}{2})$ **5.** $6, (\frac{3}{4}, \frac{3}{2})$ **7.** $\frac{1}{6}, (\frac{4}{7}, \frac{3}{4})$
9. $\frac{27}{2}, (\frac{8}{5}, \frac{1}{2})$ **11.** $(\frac{3}{8}, 3\pi/16)$
13. $(2a/5, 2a/5)$ if vertex is $(0, 0)$ and sides are along positive axes
15. $\frac{1}{10}, \frac{1}{16}, \frac{13}{80}$ **17.** $\frac{189}{20}, \frac{1269}{28}, \frac{1917}{35}$
19. $m = \pi^2/8$, $(\bar{x}, \bar{y}) = \left(\frac{2\pi}{3} - \frac{1}{\pi}, \frac{16}{9\pi}\right)$, $I_x = 3\pi^2/64$, $I_y = (\pi^4 - 3\pi^2)/16$, $I_0 = \pi^4/16 - 9\pi^2/64$
21. $\rho a^4/3$, $\rho a^4/3$; $a/\sqrt{3}$, $a/\sqrt{3}$
23. (a) $\frac{1}{2}$ (b) 0.375 (c) $\frac{5}{48} \approx 0.1042$
25. (b) (i) $e^{-0.2} \approx 0.8187$
(ii) $1 + e^{-1.8} - e^{-0.8} - e^{-1} \approx 0.3481$ (c) 2, 5
27. ≈0.500
29. (a) $\iint_D (k/20)[20 - \sqrt{(x - x_0)^2 + (y - y_0)^2}]\, dA$, where D is the disk with radius 10 mi centered at the center of the city
(b) $200\pi k/3 \approx 209k$, $200(\pi/2 - \frac{8}{9})k \approx 136k$, on the edge

Exercises 16.6 □ page 1042

1. $15\sqrt{26}$ **3.** $3\sqrt{14}$ **5.** $12\sin^{-1}\frac{2}{3}$
7. $(\pi/6)(17\sqrt{17} - 5\sqrt{5})$ **9.** $(2\pi/3)(2\sqrt{2} - 1)$
11. $a^2(\pi - 2)$ **13.** (a) ≈1.83 (b) ≈1.8616
15. $\frac{3}{2} + \frac{5}{8}\ln 5$ **17.** 3.3213 **21.** $(\pi/6)(101\sqrt{101} - 1)$

Exercises 16.7 □ page 1050

3. 1 **5.** $\frac{1}{3}(e^3 - 1)$ **7.** 4 **9.** $\frac{65}{28}$ **11.** $\frac{1}{10}$ **13.** $\frac{1}{12}$
15. $16\pi/3$ **17.** 8 **19.** $\frac{8}{15}$
21. (a) $\int_0^1\int_0^x\int_0^{\sqrt{1-y^2}} dz\, dy\, dx$ (b) $\frac{1}{4}\pi - \frac{1}{3}$
23.

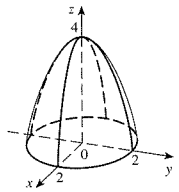


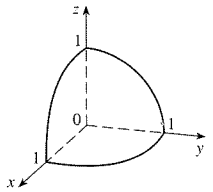
25. $\int_{-2}^2\int_0^6\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} f(x, y, z)\, dz\, dy\, dx$
$= \int_0^6\int_{-2}^2\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} f(x, y, z)\, dz\, dx\, dy$
$= \int_{-2}^2\int_0^6\int_{-\sqrt{4-z^2}}^{\sqrt{4-z^2}} f(x, y, z)\, dx\, dy\, dz$
$= \int_0^6\int_{-2}^2\int_{-\sqrt{4-z^2}}^{\sqrt{4-z^2}} f(x, y, z)\, dx\, dz\, dy$
$= \int_{-2}^2\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_0^6 f(x, y, z)\, dy\, dz\, dx$
$= \int_{-2}^2\int_{-\sqrt{4-z^2}}^{\sqrt{4-z^2}}\int_0^6 f(x, y, z)\, dy\, dx\, dz$

27. $\int_{-1}^1\int_0^{1-x^2}\int_0^y f(x, y, z)\, dz\, dy\, dx$
$= \int_0^1\int_{-\sqrt{1-y}}^{\sqrt{1-y}}\int_0^y f(x, y, z)\, dz\, dx\, dy$
$= \int_0^1\int_z^1\int_{-\sqrt{1-y}}^{\sqrt{1-y}} f(x, y, z)\, dx\, dy\, dz$
$= \int_0^1\int_0^y\int_{-\sqrt{1-y}}^{\sqrt{1-y}} f(x, y, z)\, dx\, dz\, dy$
$= \int_{-1}^1\int_0^{1-x^2}\int_z^{1-x^2} f(x, y, z)\, dy\, dz\, dx$
$= \int_0^1\int_{-\sqrt{1-z}}^{\sqrt{1-z}}\int_z^{1-x^2} f(x, y, z)\, dy\, dx\, dz$
29. $\int_0^1\int_{\sqrt{x}}^1\int_0^{1-y} f(x, y, z)\, dz\, dy\, dx = \int_0^1\int_0^{y^2}\int_0^{1-y} f(x, y, z)\, dz\, dx\, dy$
$= \int_0^1\int_0^{1-z}\int_0^{y^2} f(x, y, z)\, dx\, dy\, dz = \int_0^1\int_0^{1-y}\int_0^{y^2} f(x, y, z)\, dx\, dz\, dy$
$= \int_0^1\int_0^{1-\sqrt{x}}\int_{\sqrt{x}}^{1-z} f(x, y, z)\, dy\, dz\, dx = \int_0^1\int_0^{(1-z)^2}\int_{\sqrt{x}}^{1-z} f(x, y, z)\, dy\, dx\, dz$
31. $\int_0^1\int_0^x\int_0^y f(x, y, z)\, dz\, dy\, dx = \int_0^1\int_z^1\int_y^1 f(x, y, z)\, dx\, dy\, dz$
$= \int_0^1\int_0^y\int_y^1 f(x, y, z)\, dx\, dz\, dy = \int_0^1\int_0^x\int_z^x f(x, y, z)\, dy\, dz\, dx$
$= \int_0^1\int_z^1\int_z^x f(x, y, z)\, dy\, dx\, dz$
33. $\frac{79}{30}, (\frac{358}{553}, \frac{33}{79}, \frac{571}{553})$ **35.** a^5, $(7a/12, 7a/12, 7a/12)$
37. (a) $m = \int_{-1}^1\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}}\int_{4y^2+4z^2}^4 (x^2 + y^2 + z^2)\, dx\, dz\, dy$
(b) $(\bar{x}, \bar{y}, \bar{z})$, where
$\bar{x} = (1/m)\int_{-1}^1\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}}\int_{4y^2+4z^2}^4 x(x^2 + y^2 + z^2)\, dx\, dz\, dy$
$\bar{y} = (1/m)\int_{-1}^1\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}}\int_{4y^2+4z^2}^4 y(x^2 + y^2 + z^2)\, dx\, dz\, dy$
$\bar{z} = (1/m)\int_{-1}^1\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}}\int_{4y^2+4z^2}^4 z(x^2 + y^2 + z^2)\, dx\, dz\, dy$
(c) $\int_{-1}^1\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}}\int_{4y^2+4z^2}^4 (x^2 + y^2)(x^2 + y^2 + z^2)\, dx\, dz\, dy$
39. (a) $\frac{3}{32}\pi + \frac{11}{24}$ (b) $(\bar{x}, \bar{y}, \bar{z})$, where $\bar{x} = 28/(9\pi + 44)$, $\bar{y} = 2(15\pi + 64)/[5(9\pi + 44)]$, $\bar{z} = (45\pi + 208)/[15(9\pi + 44)]$
(c) $(68 + 15\pi)/240$
41. $I_x = I_y = I_z = \frac{2}{3}kL^5$ **43.** (a) $\frac{1}{8}$ (b) $\frac{1}{64}$ (c) $\frac{1}{5760}$
45. $L^3/8$
47. The region bounded by the ellipsoid $x^2 + 2y^2 + 3z^2 = 1$

Exercises 16.8 □ page 1057

1. 8π **3.** $\pi/6$





5. $\int_0^{\pi/2}\int_0^3\int_0^2 f(r\cos\theta, r\sin\theta, z)\, r\, dz\, dr\, d\theta$ **7.** 384π
9. 0 **11.** $2\pi/5$ **13.** 162π **15.** $\pi Ka^2/8$, $(0, 0, 2a/3)$
17. $4\pi/5$ **19.** $15\pi/16$ **21.** $4\pi(2 - \sqrt{3})$ **23.** 10π
25. (0, 0, 2.1) **27.** (a) $(0, 0, \frac{3}{8}a)$ (b) $4K\pi a^5/15$
29. $(2\pi/3)[1 - (1/\sqrt{2})]$, $(0, 0, 3/[8(2 - \sqrt{2})])$ **31.** $5\pi/6$
33. $8\pi/35$ **35.** $243\pi/5$ **37.** $136\pi/99$

Exercises 16.9 □ page 1067

1. -14 **3.** 0 **5.** $2uvw$
7. The parallelogram with vertices (0, 0), (6, 3), (12, 1), (6, −2)
9. The region bounded by the line $y = 1$, the y-axis, and $y = \sqrt{x}$
11. $\frac{11}{3}$ **13.** 6π **15.** $2\ln 3$
17. (a) $\frac{4}{3}\pi abc$ (b) 1.083×10^{12} km^3 **19.** $-\frac{66}{125}$
21. $\frac{3}{2}\sin 1$ **23.** $e - e^{-1}$

Chapter 16 Review □ page 1069

True-False Quiz

1. True **3.** True **5.** False

Exercises

1. ≈ 64.0 **3.** $4e^2 - 4e + 3$ **5.** $\frac{1}{2}\sin 1$ **7.** $\frac{2}{3}$

9. $\int_0^{\pi}\int_2^4 f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta$

11. The region outside the circle $r = 1$ and inside the cardioid $r = 1 + \sin\theta$

13. $(e - 1)/2$ **15.** $\ln\frac{3}{2}$ **17.** $\frac{1}{40}$ **19.** $\frac{41}{30}$ **21.** $81\pi/5$

23. $\frac{32}{3}$ **25.** $\pi/96$ **27.** $\frac{64}{15}$ **29.** 176 **31.** $\frac{2}{3}$

33. $2ma^3/9$

35. (a) $\frac{1}{4}$ (b) $\left(\frac{1}{3}, \frac{8}{15}\right)$

(c) $I_x = \frac{1}{12}, I_y = \frac{1}{24}; \bar{\bar{y}} = 1/\sqrt{3}, \bar{\bar{x}} = 1/\sqrt{6}$

37. (a) $(0, 0, h/4)$ (b) $\pi a^4 h/10$ **39.** $\ln(\sqrt{2} + \sqrt{3}) + \sqrt{2}/3$

41. $(\pi/8)\ln 5$ **43.** 0.0512 **45.** (a) $\frac{1}{15}$ (b) $\frac{1}{3}$ (c) $\frac{1}{45}$

47. $\int_0^1\int_0^{1-z}\int_{-\sqrt{y}}^{\sqrt{y}} f(x, y, z)\, dx\, dy\, dz$ **49.** $-\ln 2$ **51.** 0

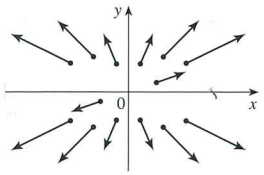
Focus on Problem Solving □ page 1072

1. 30 **3.** $\frac{1}{2}\sin 1$ **7.** (b) 0.90

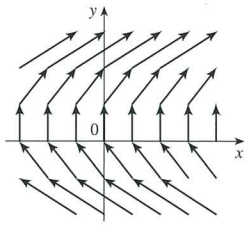
Chapter 17

Exercises 17.1 □ page 1080

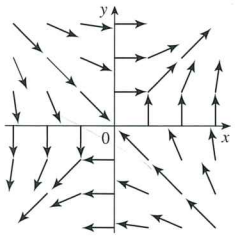
1.



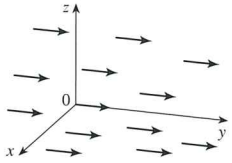
3.



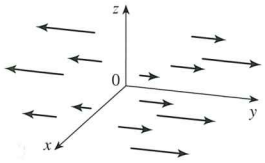
5.



7.

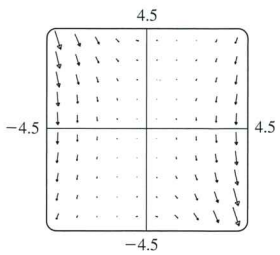


9.



11. III **13.** II
15. IV **17.** III

19.



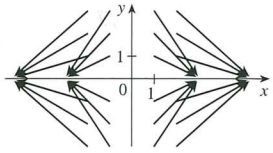
The line $y = 2x$

21. $\nabla f(x, y) = \dfrac{1}{x + 2y}\mathbf{i} + \dfrac{2}{x + 2y}\mathbf{j}$

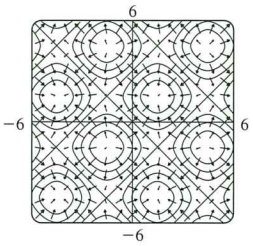
23. $\nabla f(x, y, z) = \dfrac{x}{\sqrt{x^2 + y^2 + z^2}}\mathbf{i}$

$+ \dfrac{y}{\sqrt{x^2 + y^2 + z^2}}\mathbf{j} + \dfrac{z}{\sqrt{x^2 + y^2 + z^2}}\mathbf{k}$

25. $\nabla f(x, y) = 2x\,\mathbf{i} - y\,\mathbf{j}$

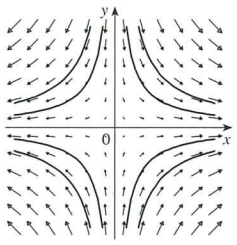


27.



29. IV **31.** II

33. (a)



(b) $y = 1/x, x > 0$

$y = C/x$

Exercises 17.2 □ page 1091

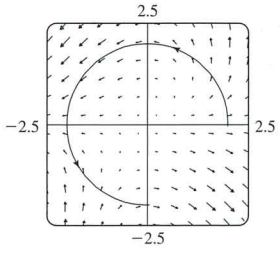
1. $\left(17\sqrt{17} - 1\right)/12$ **3.** 1638.4 **5.** $\frac{464}{5} + 9\ln 3$ **7.** $\frac{17}{3}$

9. 320 **11.** $\sqrt{14}(e^6 - 1)/12$ **13.** $\frac{16}{11}$ **15.** $\frac{77}{6}$

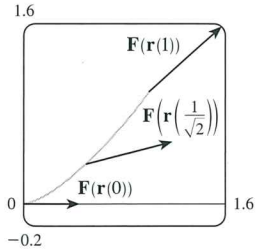
17. (a) Positive (b) Negative **19.** $-\frac{59}{105}$

21. $\frac{6}{5} - \cos 1 - \sin 1$

23. $3\pi + \frac{2}{3}$



25. (a) $\frac{11}{8} - 1/e$ (b)



27. $\frac{945}{16,777,216}\pi$ **29.** 0.052 **31.** $2\pi k, (4/\pi, 0)$
33. (a) $\bar{x} = (1/m)\iint_C x\rho(x, y, z)\, ds$,
$\bar{y} = (1/m)\int_C y\rho(x, y, z)\, ds$,
$\bar{z} = (1/m)\int_C z\rho(x, y, z)\, ds$, where $m = \int_C \rho(x, y, z)\, ds$
(b) $2\sqrt{13}\,k\pi, (0, 0, 3\pi)$
35. $I_x = k\left((\pi/2) - \frac{4}{3}\right), I_y = k\left((\pi/2) - \frac{2}{3}\right)$
37. $2\pi^2$ **39.** $\frac{23}{88}$ **41.** 1.67×10^4 ft-lb **43.** ≈22 J

Exercises 17.3 □ page 1100

1. 40 **3.** $f(x, y) = 3x^2 + 5xy + 2y^2 + K$
5. Not conservative **7.** $f(x, y) = x^2 \cos y - y \sin x + K$
9. $f(x, y) = ye^x + x \sin y + K$ **11.** (b) 16
13. (a) $f(x, y) = \frac{1}{4}x^4y^4$ (b) 4
15. (a) $f(x, y, z) = xy + yz$ (b) 15
17. (a) $f(x, y, z) = x^2z + x \sin y$ (b) 2π
19. $25 \sin 1 - 1$ **21.** $\frac{8}{3}$ **23.** No **25.** No
29. (a) Yes (b) Yes (c) Yes
31. (a) Yes (b) Yes (c) No

Exercises 17.4 □ page 1108

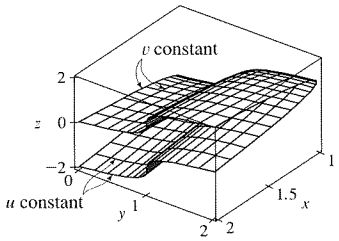
1. 6 **3.** $\frac{2}{3}$ **7.** $e - 1$ **9.** $\frac{1}{3}$ **11.** -24π **13.** 0
15. $\pi + \frac{16}{3}\left[\left(1/\sqrt{2}\right) - 1\right]$ **17.** $-\frac{1}{12}$ **19.** $3\pi/8$
21. (c) $\frac{9}{2}$ **23.** $\left(\frac{1}{3}, \frac{1}{3}\right)$

Exercises 17.5 □ page 1115

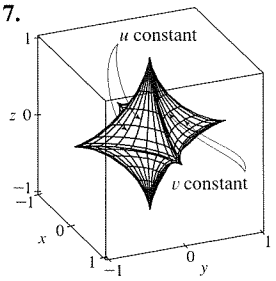
1. (a) $-y\,\mathbf{i} - z\,\mathbf{j} - x\,\mathbf{k}$ (b) $x + y + z$
3. (a) $-x^2\,\mathbf{i} + 3xy\,\mathbf{j} - xz\,\mathbf{k}$ (b) yz **5.** (a) $\mathbf{0}$ (b) 1
7. (a) $\frac{y}{z^2}\mathbf{i} - \frac{x}{z^2}\mathbf{j}$ (b) $\frac{2z + 1}{z^2}$
9. (a) Negative (b) curl $\mathbf{F} = \mathbf{0}$
11. (a) Zero (b) curl $\mathbf{F}$ points in the negative z-direction
13. $f(x, y, z) = xyz + K$ **15.** $f(x, y, z) = x^2y + y^2z + K$
17. Not conservative **19.** No

Exercises 17.6 □ page 1125

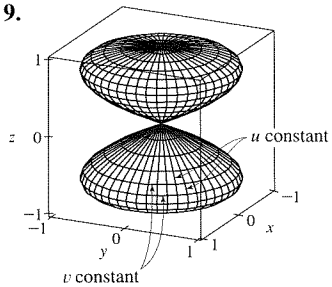
1. Circular paraboloid with axis the z-axis
3. Circular cylinder with axis the x-axis
5.



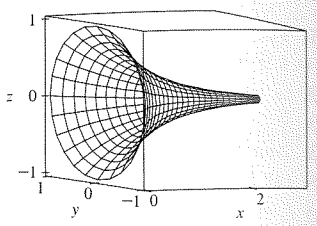
7.



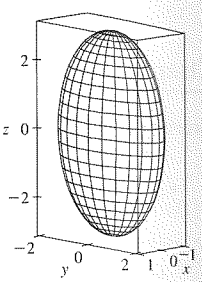
9.



11. IV **13.** I **15.** II
17. $x = 1 + u + v, y = 2 + u - v, z = -3 - u + v$
19. $x = x, y = 6 - 3x^2 - 2z^2, z = z, 3x^2 + 2z^2 \leq 6$
21. $x = 2 \sin\phi \cos\theta, y = 2 \sin\phi \sin\theta,$
$z = 2 \cos\phi, 0 \leq \phi \leq \pi/4, 0 \leq \theta \leq 2\pi$
23. $x = r \cos\theta, y = r \sin\theta, z = 5, 0 \leq r \leq 4, 0 \leq \theta \leq 2\pi$
[or $x = x, y = y, z = 5, x^2 + y^2 \leq 16$]
25. $x = x, y = e^{-x} \cos\theta,$
$z = e^{-x} \sin\theta, 0 \leq x \leq 3,$
$0 \leq \theta \leq 2\pi$



27. (a) Direction reverses (b) Number of coils doubles
29. $3x - y + 3z = 3$ **31.** $x = 0$
33. $4\sqrt{6}\,\pi$ **35.** $(\pi/6)\left(17\sqrt{17} - 5\sqrt{5}\right)$
37. $\left(\sqrt{21}/2\right) + \frac{17}{4}\left[\ln\left(2 + \sqrt{21}\right) - \ln\sqrt{17}\right]$
39. $2a^2(\pi - 2)$ **41.** $\pi\left(2\sqrt{6} - \frac{8}{3}\right)$
43. (a) ≈1.83 (b) ≈1.8616 **45.** (b)



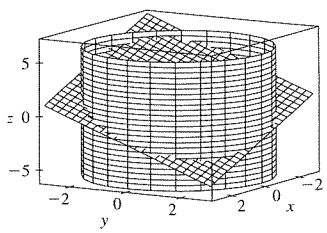
(c) $\int_0^{2\pi}\int_0^{\pi} \sqrt{36 \sin^4 u \cos^2 v + 9 \sin^4 u \sin^2 v + 4 \cos^2 u \sin^2 u}\, du\, dv$
47. 4.4506

Exercises 17.7 □ page 1137

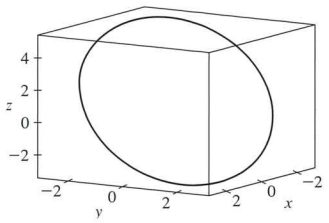
1. $8\left(1 + \sqrt{2} + \sqrt{3}\right) \approx 33.17$ **3.** 900π
5. $171\sqrt{14}$ **7.** $\sqrt{3}/24$ **9.** $\left(33\sqrt{33} - 17\sqrt{17}\right)/6$
11. $\pi\sqrt{2}/4$ **13.** 16π **15.** 16π **17.** 0 **19.** $\frac{713}{180}$
21. $-\frac{1}{6}$ **23.** 108π **25.** 0 **27.** 48 **29.** 0.1642
31. 3.4895
33. $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D [P(\partial h/\partial x) - Q + R(\partial h/\partial z)]\, dA$,
where D = projection on xz-plane
35. $(0, 0, a/2)$
37. (a) $I_z = \iint_S (x^2 + y^2)\rho(x, y, z)\, dS$
(b) $4329\sqrt{2}\,\pi/5$
39. $194{,}400\pi$ **41.** $8\pi a^3\varepsilon_0/3$ **43.** 1248π

Exercises 17.8 □ page 1143

3. 0 **5.** 0 **7.** -1 **9.** -4π
11. (a) $81\pi/2$ (b)



(c) $x = 3\cos t$, $y = 3\sin t$, $z = 1 - 3(\cos t + \sin t)$, $0 \le t \le 2\pi$



17. 16

Exercises 17.9 □ page 1150

1. Negative at P_1, positive at P_2 **7.** 8 **9.** 0
11. $9\pi/2$ **13.** $12\pi/5$ **15.** $-81\pi/2$
17. $341\sqrt{2}/60 + \frac{81}{20}\arcsin(\sqrt{3}/3)$ **19.** $13\pi/20$

Chapter 17 Review □ page 1153

True-False Quiz

1. False **3.** True **5.** False **7.** True

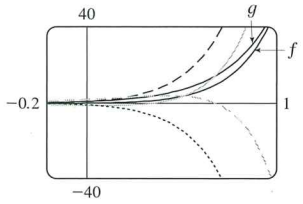
Exercises

1. (a) Negative (b) Positive **3.** $4\sqrt{5}$ **5.** $-\pi$ **7.** $\frac{17}{2}$
9. 5 **11.** $f(x, y) = x\sin y - \cos y$ **13.** π^2 **17.** -8π
25. $\frac{1}{6}(27 - 5\sqrt{5})$ **27.** $\pi(391\sqrt{17} + 1)/60$
29. $-64\pi/3$ **33.** $-\frac{1}{2}$ **37.** -4 **39.** 21

Chapter 18

Exercises 18.1 □ page 1165

1. $y = c_1e^{4x} + c_2e^{2x}$ **3.** $y = e^{-4x}(c_1\cos 5x + c_2\sin 5x)$
5. $y = c_1e^{x} + c_2xe^{x}$ **7.** $y = c_1\cos(x/2) + c_2\sin(x/2)$
9. $y = c_1 + c_2e^{-x/4}$ **11.** $y = c_1e^{(1+\sqrt{2})t} + c_2e^{(1-\sqrt{2})t}$
13. $y = e^{-t/2}\left[c_1\cos(\sqrt{3}t/2) + c_2\sin(\sqrt{3}t/2)\right]$
15.



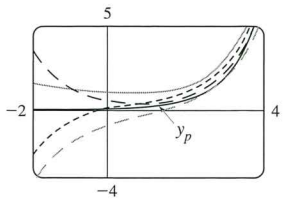
All solutions approach 0 as $x \to -\infty$ and approach $\pm\infty$ as $x \to \infty$.

17. $y = 2e^{-3x/2} + e^{-x}$ **19.** $y = e^{x}(\cos x + \sin x)$
21. $y = 2e^{1-x} + e^{3(x-1)}$ **23.** $y = -\frac{1}{3}\sin 3x$
25. $y = 3xe^{-2x+2}$ **27.** No solution
29. $y = \dfrac{e^5}{e^6 - 1}e^{-x} + \dfrac{e^2}{1 - e^6}e^{2x}$
31. $y = e^{-2x}(2\cos 3x - e^{\pi}\sin 3x)$
33. (b) $\lambda = n^2\pi^2/L^2$, n a positive integer; $y = C\sin(n\pi x/L)$

Exercises 18.2 □ page 1172

1. $y = c_1e^{-2x} + c_2e^{-x} + \frac{1}{2}x^2 - \frac{3}{2}x + \frac{7}{4}$
3. $y = c_1 + c_2e^{2x} + \frac{1}{40}\cos 4x - \frac{1}{20}\sin 4x$
5. $y = e^{2x}(c_1\cos x + c_2\sin x) + \frac{1}{10}e^{-x}$
7. $y = \frac{3}{2}\cos x + \frac{11}{2}\sin x + \frac{1}{2}e^{x} + x^3 - 6x$
9. $y = \frac{5}{8}e^{x} - \frac{17}{32}e^{-x} + e^{3x}\left[\frac{1}{8}x - \frac{3}{32}\right]$

11.

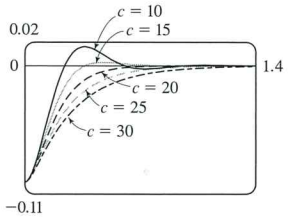


The solutions are all asymptotic to $y_p = e^{x}/10$ as $x \to \infty$. Except for y_p, all solutions approach either ∞ or $-\infty$ as $x \to -\infty$.

13. $y_p = (Ax^4 + Bx^3 + Cx^2 + Dx + E)e^{2x}$
15. $y_p = xe^{x}(A\cos x + B\sin x)$
17. $y = c_1\cos 2x + c_2\sin 2x + \frac{1}{4}x$
19. $y = c_1e^{x} + c_2xe^{x} + e^{2x}$
21. $y = (c_1 + x)\sin x + (c_2 + \ln\cos x)\cos x$
23. $y = [c_1 + \ln(1 + e^{-x})]e^{x} + [c_2 - e^{-x} + \ln(1 + e^{-x})]e^{2x}$
25. $y = \left[c_1 - \frac{1}{2}\int (e^{x}/x)\,dx\right]e^{-x} + \left[c_2 + \frac{1}{2}\int (e^{-x}/x)\,dx\right]e^{x}$

Exercises 18.3 □ page 1179

1. $x = 0.36\sin(10t/3)$ **3.** $x = -\frac{1}{5}e^{-6t} + \frac{6}{5}e^{-t}$ **5.** $\frac{49}{12}$ kg
7.



11. $Q(t) = (-e^{-10t}/250)(6\cos 20t + 3\sin 20t) + \frac{3}{125}$, $I(t) = \frac{3}{5}e^{-10t}\sin 20t$
13. $Q(t) = e^{-10t}\left[\frac{3}{250}\cos 20t - \frac{3}{500}\sin 20t\right] - \frac{3}{250}\cos 10t + \frac{3}{125}\sin 10t$

Exercises 18.4 □ page 1184

1. $c_0\sum_{n=0}^{\infty}\dfrac{x^n}{n!} = c_0e^{x}$ **3.** $c_0\sum_{n=0}^{\infty}\dfrac{x^{3n}}{3^n n!} = c_0e^{x^3/3}$
5. $c_0\sum_{n=0}^{\infty}\dfrac{(-1)^n}{2^n n!}x^{2n} + c_1\sum_{n=0}^{\infty}\dfrac{(-2)^n n!}{(2n+1)!}x^{2n+1}$
7. $c_0 + c_1x + c_0\dfrac{x^2}{2} + c_0\sum_{n=2}^{\infty}\dfrac{(-1)^{n-1}(2n-3)!}{2^{2n-2}n!(n-2)!}x^{2n}$
9. $\sum_{n=0}^{\infty}\dfrac{x^{2n}}{2^n n!} = e^{x^2/2}$
11. $x + \sum_{n=1}^{\infty}\dfrac{(-1)^n 2^2 5^2 \cdot \cdots \cdot (3n-1)^2}{(3n+1)!}x^{3n+1}$

Chapter 18 Review □ page 1185

True-False Quiz

1. True **3.** True

Exercises

1. $y = c_1e^{5x} + c_2e^{-3x}$ **3.** $y = c_1\cos(\sqrt{3}x) + c_2\sin(\sqrt{3}x)$
5. $y = e^{-x}(c_1 + c_2x) - \frac{3}{50}\cos 3x - \frac{2}{25}\sin 3x$
7. $y = c_1\cos(3x/2) + c_2\sin(3x/2) + \frac{2}{9}x^2 - \frac{43}{81}$
9. $y = c_1e^{x} + c_2e^{2x} + xe^{2x}$ **11.** $y = 5 - 2e^{-6(x-1)}$
13. $y = (e^{4x} - e^{x})/3$ **15.** $\sum_{n=0}^{\infty}\dfrac{(-2)^n n!}{(2n+1)!}x^{2n+1}$
17. $Q(t) = -0.02e^{-10t}(\cos 10t + \sin 10t) + 0.03$
19. (c) $2\pi/k \approx 85$ min (d) $\approx$17,600 mi/h